Bidding Away the Incentives to Drill Revenue-Sharing Auctions in Mexico's Oil Industry

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Abstract

Many governments allocate oil fields and mines through revenue-sharing auctions, which award the resource to the firm that bids the highest share of its future revenue. Because the winning bid acts as a tax on the winner's later revenue, rules that raise it, such as a higher minimum bid, can discourage the drilling on which the government's revenue depends. Using the bids and subsequent drilling of the 104 blocks that Mexico awarded in its 2015–2018 oil auctions, and instrumenting the winning bid with the competitiveness of the other blocks in the same tender to address its endogeneity, I find that a 10-percentage-point higher bid lowers the hazard of starting to drill by 30 percent and delays the start of drilling. A model of bidding and the timing of drilling separates this incentive effect from the selection of lower-cost winners. Raising the minimum bid above about 20 percent lowers the government's revenue because the incentive effect dominates. Bonus and profit-sharing auctions avoid the distortion but are hard to implement where insurance markets are thin and costs are hard to monitor. When the outcome of an auction shapes the winner's later decisions, the government should design the auction with the winner's incentives in mind.

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1 Introduction

Governments lease oil fields to private companies and ask for part of the oil's value in return. A government wants to raise public revenue from the contractor, and at the same time it wants the field to go to a contractor that will develop it well. However, it does not know each firm's costs in advance, so it cannot simply pick the firm that would develop the field most cheaply. A common response to this problem is the revenue-sharing auction. Each firm bids the share of the field's future revenue that it will pay to the government, and the government leases the field to the firm that offers the highest share. Firms thus compete over a tax rate on their own revenue. After the auction the winner drills the field, produces oil or gas, and pays the government its winning share of revenue until the lease ends.

A government that runs such an auction is tempted to design it to raise the winning bid, for example by raising the minimum bid. What the government collects, however, depends on what the winner does after the auction, in particular on whether and when it explores and drills the field. The winner bears the full cost of drilling but keeps only the remaining fraction of the revenue it produces, so the winning share acts as a tax on the return to its drilling investment, and a higher share weakens its incentive to invest. A policy that raises the winning share therefore does not always raise the government's revenue, because the higher share can discourage the winner from investing in drilling. This paper studies how the design of revenue-sharing auctions affects the winner's drilling decisions and the government's objectives.

The question matters for resource-rich developing countries, which use revenue-sharing widely, as in the oil industries of Mexico and Brazil and in coal mining in India. The alternatives that would not distort exploration are often unavailable to them. A profit-sharing auction requires the winner to report its costs, and that report is neither incentive compatible nor easy for a government to audit. A bonus auction, in which the winner pays a fixed sum up front, places the entire risk of uncertain future production on the bidder, which pays regardless of what the field yields. Where financial markets are imperfect, that risk deters entry. The revenue-sharing auction is therefore often the only feasible way to allocate a field in emerging markets. Because oil revenue accounts for a large share of public finances in these countries, the auction's effect on drilling and government revenue matters for the public budget. The same structure also appears outside resource markets. Renewable-energy procurement sometimes uses pay-as-produced power purchase auctions, in which a buyer of electricity looks for a cheap supplier. Potential suppliers bid a price per unit of electricity, the buyer contracts with the supplier that offers the lowest price, and the supplier then decides how much to generate each hour at that price. As in a revenue-sharing auction, the low price that competition delivers can discourage the winner from producing. In both settings the design of the auction has to account for what the winner does after the auction as well as for how firms bid.

As an empirical case, I study the revenue-sharing auctions of Mexico's upstream oil sector. Between 2015 and 2018 the government awarded 104 oil blocks by revenue-sharing auctions. I link every bid to the subsequent drilling, investment and production of each awarded block. I first show how the winning bid relates to the winner's later drilling and production. I instrument the winning bid with the floors and ceilings the government set on the other blocks offered in the same tender, which shift the competition for a block without depending on its own characteristics. A 10-percentage-point higher winning bid lowers the hazard of starting to drill by 30 percent and delays the first drilling by 1.2 years. It also raises the government's revenue from the contract. These estimates combine two effects. A higher share lowers the winner's return on investment and so discourages drilling. A higher bid also comes from a winner with a lower cost, which is more likely to drill. The reduced form cannot separate these incentive and selection channels.

I then estimate a structural model that separates the two channels and lets me compare allocation mechanisms that were never used. In the model, firms differ in their drilling costs, and each firm's cost is its private information, unknown to its rivals and to the government. The firms make two decisions. In the auction each firm bids a revenue share, weighing its chance of winning against the value of the lease at the current oil price. After the auction the winner decides each month whether to start drilling, given its drilling cost and its expectation of the oil price. It drills once the price is high enough relative to its cost, and otherwise it waits another month in case the price rises. I estimate the value of the right to drill, the equilibrium bidding strategy and the distribution of drilling costs from the observed bids and the timing of drilling.

The estimates give two main results. First, a counterfactual that varies the minimum bid shows the cost of the incentive effect. At low floors a higher minimum bid raises revenue while discouraging drilling, because the higher tax rate outweighs the loss in drilling. Above a floor of about 20 percent the incentive effect dominates, and raising the floor lowers the government's revenue. Once the winner's drilling decision is taken into account, the government cannot simply raise the minimum bid to improve its outcomes. Second, I compare the revenue-sharing auction with two alternatives, a first-price auction in which firms bid an upfront bonus and an auction in which firms bid a share of profit instead of revenue. Neither distorts the winner's drilling decision. Onshore, the winner drills within the period on 76 percent of awarded blocks under both, against 61 percent in the actual auction, and total surplus rises from $107 million to $110 million per offered block. The two alternatives divide that surplus very differently. The bonus auction favors the winner, and the government collects $37 million against $53 million in the actual auction, because bidders can extract a larger information rent from their private cost information. The profit-sharing auction favors the government, which collects $89 million. Both raise total surplus, but the risk-sharing and cost-monitoring problems described above make them hard to implement in emerging markets.

This paper contributes to a growing literature on how auction design affects the investment and production decisions that follow the auction. To my knowledge, it is the first to study the incentive effect of revenue-sharing auctions, in which bidders compete on the share of revenue itself, and to show two of its consequences. First, the auction can be inefficient ex post. Second, a rule that improves the outcome of the auction itself, such as a higher minimum bid that raises the winning bid, can lower the government's final revenue, because the higher share discourages the winner from drilling. Neither consequence arises in the formats that earlier papers study, because in those formats bidders do not compete on a revenue share. Nguyen (2026) studies a first-price auction in which bidders compete on an upfront payment, the bonus. The paper shows that the government can encourage the winner to explore more by disclosing the losing bids after the auction, because they carry information about the uncertain reserve. A bonus does not depend on output, however, so it does not distort the winner's exploration decision, and a policy that raises the winning bid has no incentive effect in that setting. Bhattacharya et al. (2022) study how the government should set the royalty rate when the royalty discourages drilling. Their firms bid a bonus at a royalty rate that the government sets, so every bidder faces the same rate and the winner is never selected by the rate it offers. The rate is then a policy choice, not the outcome of competition that auction rules push up. Kong et al. (2022) study auctions in which firms bid both a royalty rate and an upfront payment. They show that a low-cost bidder lowers its upfront payment while raising its royalty rate, which keeps its chance of winning and reduces its expected total payment. Bidders there can trade one component of the bid against the other, whereas in a revenue-sharing auction the share is the only margin of competition. My results extend this literature on the interaction between auction design and the winner's post-auction behavior to revenue-sharing auctions, in which the bid itself sets the winner's incentive to drill.

The rest of the paper proceeds as follows. Section 2 describes Mexico's oil sector and its auctions. Section 3 presents the reduced-form evidence. Section 4 develops the model of entry, bidding and the timing of drilling, and Section 5 describes its estimation. Section 6 studies the minimum bid and compares allocation mechanisms. Section 7 concludes.

2 Revenue-Sharing Auctions in Mexico's Oil Industry

2.1 Mexico's Upstream Oil Sector

Mexico is one of the world's major oil producers. It was the eleventh largest in 2024 and the fourth largest in the Americas after the United States, Canada, and Brazil (Energy Institute, 2025). It produced an average of 1.9 million barrels per day of petroleum and other liquids in 2022, half of its 2004 peak of 3.8 million barrels per day. About half of its crude oil is sold abroad and half is refined at home. Exports of crude oil and lease condensate

were 1.16 million barrels per day in 2021, while Mexico's own refineries processed about 0.82 million barrels per day in 2022 (U.S. Energy Information Administration, 2023). Oil is found both onshore and offshore. Mexico's proved oil reserves were 6.0 billion barrels in January 2023. About 70 percent lie offshore, mostly in the Sureste Basin off the southeast coast, and 30 percent onshore, mostly in the Tampico-Misantla Basin (U.S. Energy Information Administration, 2023). The blocks in my sample lie onshore, in shallow water, and in deep water, and range from onshore fields with known reserves to frontier deepwater areas that had not yet been drilled.

2.2 Auction System

Hydrocarbons in the ground belong to the Mexican state. Until the energy reform of 2013 and 2014, only the national oil company, Pemex, could drill for and produce them. The reform let the state lease blocks to private firms as well (U.S. Energy Information Administration, 2023). The government allocates each block to a firm, or a consortium of firms, under a contract that gives it the exclusive right to drill the block and, if it finds oil, to develop and produce it. The firm pays all the costs and in return keeps the oil it produces, net of what the contract requires it to pay the state. Alicence (licencia) makes the firm pay the state a share of the value of the oil it produces, so the state shares in revenue. Their payment does not depend on the firm's costs, so the share the firm offers is a pure tax on the value it produces.

The National Hydrocarbons Commission (CNH) awards each block by auction. CNH has held nine tenders since the first was announced in 2014. Table 1 summarizes them, and Figure A1 in Appendix A maps the blocks. Round 1 consisted of four tenders between July 2015 and December 2016, Round 2 of four tenders between June 2017 and January 2018, and Round 3 of a single tender in March 2018. Across the nine tenders CNH offered 157 blocks and awarded 104. Competition varied widely. Round 1 Tender 3 drew 39 bidders and an average of 6.7 valid bids per block, while every other tender averaged fewer than four, and in Round 1 Tender 1 only 2 of the 14 blocks received a winning bid.

Each block is sold in a separate sealed-bid auction. A bid is the percentage of the block's future revenue, the value of the oil and gas it produces, that the bidder will pay to the government for the life of the contract. The block goes to the bidder that offers the government the highest percentage. Bids could also include a commitment to additional exploration work, which carried a small weight in the score. But, we just focus on the revenue share bid because the commitment carries little weight in the auctions I study and the commitments bidders submitted vary little within these rounds.

The Ministry of Finance (SHCP) set a floor on the government share for every block, so no bid below it could win, and from Round 2 on a ceiling as well. Table A2 in Appendix A reports the floors and ceilings by tender. The average floor ranges from 2 percent in Round 1

Tender 4 to 35 percent in Round 1 Tender 1. Floors also vary across blocks within a tender, but they rarely bind. Only 1 of the 104 winning bids sits at its floor. Round 1 placed no upper limit on the share, while every block in Rounds 2 and 3 had one. The average ceiling ranges from 20 percent in Round 2 Tender 4 to 75 percent in Round 2 Tender 1. Unlike floors, ceilings bind often. Of the 66 awarded blocks with a ceiling, 34 were won with a bid exactly at the ceiling, including 26 of the 40 in Round 2 Tenders 2 to 4. When more than one bidder offers the ceiling, the tie is broken by a sealed cash payment. The bidder that offers the most cash wins and pays what it offered, so among the bidders at the ceiling the auction becomes a first-price auction in cash. For these blocks the winning share is set by the ceiling, and competition moves to the cash payment. The variation in floors and ceilings across the blocks of a tender is the source of the instruments I use in Section 3.

2.3 Information

Before each tender, CNH publishes for every block a single estimate of the hydrocarbons that can be recovered from it, which is a forecast of the block's future production. For blocks offered for exploration, the estimate is the block's prospective resources in millions of barrels of oil equivalent, reported as a distribution with P90, P50, mean and P10 values. For the discovered fields of Round 1 Tenders 2 and 3, it is the original volume in place and the 1P, 2P and 3P reserves. The same figures appear on the block's technical sheet and on the fact sheet CNH released after the auction. Firms that register for a tender can also enter CNH's data room, which holds the seismic data and well records the state has on the block. Every firm in the data room sees the same data. A firm cannot learn much more on its own before the auction. Drilling a well on a block requires a contract for that block, so no bidder can drill before it wins. Firms can acquire seismic surveys before a tender, but only under a non-exclusive authorization, and survey firms sell the resulting data to any buyer, so a bidder that buys them learns what its rivals can buy as well. Every bidder therefore starts from much the same forecast of how much the block will produce, which the model in Section 4 uses when it treats the block's oil as common and known.

2.4 Exploration and Drilling

Exploration is the search for oil within the block. The winner first studies the block with seismic surveys and then drills exploratory wells to test whether the structures it has mapped hold oil. A well that finds oil is a discovery. The winner then drills further wells to measure its size, and only if it is commercial does it file a development plan and build the wells and facilities that produce it. A block cannot produce before it has been drilled, so drilling decides whether and when the block yields any revenue for the firm or for the state.

Exploration is costly. The exploration investment in the winners' plans averages $48

million per contract onshore, $103 million in shallow water, and $145 million in deep water, where a single exploratory well can cost far more than a whole onshore program. Its returns are uncertain and arrive late. Of the 76 exploration contracts in my sample, every one filed an exploration plan, but by January 2025 only 14 had produced, 12 of them onshore and none in deep water. The contract sets a deadline for exploration but leaves its timing within that deadline to the winner. For instance, in the onshore licences of Round 2, the winner must file an exploration plan within 180 days of the contract's start. What one winner learns from drilling also has value beyond its own block. All technical information the winner gathers, including seismic data, well logs and rock samples, belongs to the state and must be delivered to CNH, which may use it or sell it (Comisión Nacional de Hidrocarburos, 2017). A well that finds oil, or fails to, therefore tells the government and other firms about the geology of neighboring blocks. The winner does not capture this value, which is one reason the state cares about the timing of drilling beyond the revenue it collects from the block itself.

The winner signs a contract whose length depends on the block. Fields that had already been discovered, offered in Round 1 Tenders 2 and 3, were awarded under 25-year extraction contracts with no exploration phase. Blocks that still had to be drilled were awarded under exploration-and-extraction contracts, which last 30 years in shallow water and onshore and 35 years in deep water.

2.5 Data

I combine five public sources, all published by the Mexican government or its agencies, and link them at the level of the block. Table A1 in Appendix A reports summary statistics for every variable I use. The bid data come from CNH's records of the nine tenders, published on the government's Rondas México website. For each of the 157 blocks offered, CNH released a one-page result fact sheet (resultado de la licitación) that lists every valid bid, winning and losing, together with the bidder, the government share it offered, its additional work commitment, and the block's floor and ceiling. From these sheets I recover 369 valid bids on 108 blocks, the winning bid on each of the 104 awarded blocks, and, for blocks won at the ceiling, the cash payments that broke the tie. The same website lists the firms prequalified for each tender and, for every consortium, its member companies and their countries of origin. I use this roster to count the potential entrants in each tender and to identify each bidder's members.

Block characteristics come from the technical sheet (ficha técnica) that CNH published for every block before each tender. The sheet describes the block's location, area, geological province and main hydrocarbon. For blocks offered for exploration, the sheet also reports CNH's estimate of prospective resources, which I use at the median (P50). I recover this estimate for 74 of the 76 exploration blocks. For Round 2 Tender 4 and Round 3 Tender 1, CNH reported prospective resources for groups of neighboring blocks rather than for each

block, so the estimate is constant within each group. The contract term separates two kinds of block. The 28 blocks of Round 1 Tenders 2 and 3 cover fields already discovered and were awarded under 25-year extraction contracts. The other 76 still had to be drilled and were awarded under exploration-and-extraction contracts of 30 or 35 years.

Post-auction activity comes from the Hydrocarbon Information System (SIH), the public database that CNH maintains for the upstream sector. SIH records, for each contract, its investment plans by year and type, separating plans to drill the block, to measure a discovery and to develop it. It also records monthly production of liquids and gas by contract and production by well, which assigns each producing well to a contract. I observe these series through January 2025. I measure exploration by the year of the first exploration plan, the planned exploration investment, the number of years with an exploration plan, the year of the first plan to measure a discovery, and the number of wells assigned to the contract. In the structural model, starting to drill means committing the large investment in the block, and I date it by the first plan to measure a discovery. Every exploration contract files an exploration plan early, so that plan carries little information on when the winner commits. I measure production by whether the block has produced and by the month of first positive output. Because SIH identifies contracts by their CNH contract number, these series link to the auction records directly.

Government revenue comes from the Mexican Petroleum Fund (FMP), the trust that receives every payment that upstream contracts make to the state and publishes these payments by contract. For licences I sum the base and additional royalties paid in dollars. For production-sharing contracts, which pay the state's share in kind, I value the oil, condensate and gas delivered at the monthly contract prices that the FMP reports. The FMP identifies contracts by its own registry number and names them after their fields, so I link them to the CNH contracts through the field names in the bid records and the names of the wells SIH assigns to each contract. Blocks that never produced paid nothing and enter with zero revenue. Eleven blocks that produced have no FMP record and are left missing. For oil prices, I take daily Brent and Mexican export blend prices from SIH and deflate them to December 2024 dollars with the US consumer price index for all urban consumers (CPI-U) from the Federal Reserve Bank of St. Louis. The price process in Section 5.1 uses the end-of-month real Brent price.

The reduced-form analysis uses the 104 awarded blocks. The winning bid averages 39.6 percent of the contract's value, with a standard deviation of 22.4 percentage points, and ranges from 3.9 to 85.7 percent. By January 2025, 76 blocks had an exploration plan and 38 had produced. The structural model describes licences, in which the winner pays its share on the value of output and bears the whole cost, so I estimate it on the 63 exploration-and-extraction licences offered in Round 1 Tender 4 and Round 2 Tenders 2 to 4, of which 48 were awarded.

Figure 1 shows that drilling in Mexico responds to the oil futures price, which reflects

Table 1—Auctions, Bidders and Bids, by Tender

Round	Tender	Year	Month	Auctions	Bidders	Mean active bidders	Bids (%)			Mean winning share (%)
							Min	Max	Mean	
1	1	2015	Jul	14	4	0.4	55	69	61	62
1	2	2015	Sep	5	9	3.0	47	84	63	76
1	3	2015	Dec	25	39	6.7	2	86	31	55
1	4	2016	Dec	10	10	1.3	4	27	13	14
2	1	2017	Jun	15	20	1.9	20	75	55	57
2	2	2017	Jul	10	5	1.2	4	45	19	24
2	3	2017	Jul	14	17	3.7	4	45	32	35
2	4	2018	Jan	29	19	1.3	5	20	14	16
3	1	2018	Mar	35	18	1.0	24	65	47	46
All				157	–	2.4	2	86	34	40

Notes: Each row is one tender. Auctions count the blocks offered. Bidders counts distinct bidders, a consortium as one, with at least one valid bid, and mean active bidders is valid bids per auction. Bid statistics are over valid bids, and the mean winning share is over awarded blocks. Source: CNH result fact sheets.

what the market expects oil to be worth in the coming months. It plots the number of drilling rigs active each month across all operators in Mexico against the real futures price from 2000 to 2024. The two series rise and fall together, with a correlation of 0.60. Rigs climbed above 100 during the price boom of the 2000s and stayed there until 2014. When the futures price fell from $137 to $67 between mid-2014 and mid-2016, the number of active rigs fell from about 100 to 18. It recovered to about 50 by 2019 as the price rose again. This pattern suggests that firms drill more when they expect oil to be worth more and hold back when they expect it to be worth less. The trigger rule of the model in Section 4 produces the same pattern.

3 Effects of the Winning Bid on Drilling and Production

This section asks how the winning bid affects the winner's later drilling and production decisions. Because the bid is the government's share of the contract's value, a higher winning bid lowers the return the winner keeps on every barrel it later finds and produces. The contract gives the winner the right to drill and produce in the block, and much of its value lies in the option to choose when to use that right. Beyond filing an exploration plan and completing a minimum work program within the initial exploration period, the contract leaves the timing of investment to the winner. A winner starts drilling when the discounted return from drilling exceeds its cost and otherwise waits. Winners are able to begin soon after the contract takes effect, yet many delay. Because a higher government share lowers the return from drilling, I expect winners with higher bids to wait longer and to invest less,

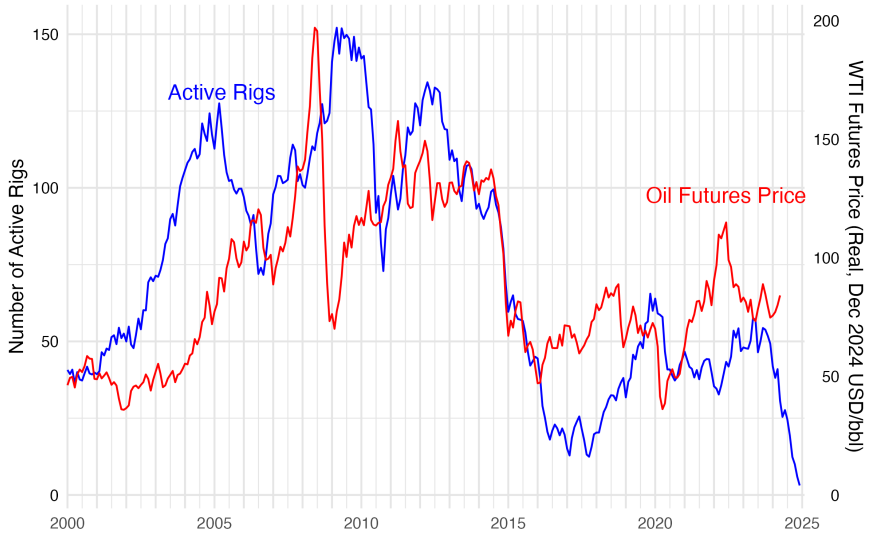


Figure 1: Active Drilling Rigs in Mexico and the Oil Price, 2000–2024

Notes: Monthly number of active drilling rigs summed over all operators, from SIH, and the monthly mean NYMEX WTI futures price for the fourth-nearest contract, from EIA (through April 2024), in December 2024 US dollars, deflated with the CPI-U.

other things equal.

3.1 Empirical Strategy

I measure the winner's response with six block-level outcomes. On the exploration side I observe the number of years from the contract year to the first drilling, the total exploration investment in millions of nominal US dollars, the number of years in which the winner drills, and the number of wells the winner drilled. On the production side I observe whether the block has ever reported positive output during the lease period and the number of months to first production. I estimate exploration investment in logs on the 76 blocks with positive planned investment, an intensive-margin measure.

For block i auctioned in tender t, I estimate $y_i = \alpha + \beta\, b_i + \varepsilon_i$, where y_i is one of the outcomes above and b_i is the winning bid, the government share in percent. In the duration models for the time to first drilling or first production, β enters the hazard as $h_i(\tau) = h_0(\tau)\exp(\beta b_i)$. The coefficient β is a log hazard ratio, so a one-percentage-point increase in the winning bid multiplies the probability that a block that has not yet drilled begins drilling in a given period by $\exp(\beta)$. A positive coefficient therefore means a shorter time to the event. The times to first drilling and first production are right-censored because some blocks had neither drilled nor produced by the end of the data. The data end in January 2025, and by then 28 of the 104 blocks had not drilled and 66 had not produced, so for these blocks I know only that the duration exceeds the time elapsed since the start of the contract year, 10 to 11 years for drilling and 84 to 120 months for production. Dropping the censored

blocks keeps only blocks that reached the event within the observation window, which selects the sample on the outcome. Recording the censoring time as the duration treats a block that has not yet drilled as one that drilled on the last day of the data, which understates the longest durations. The Cox model avoids both problems. Each censored block stays in the risk set until it is censored and contributes the information that its event had not occurred by then. For comparison, column (2) reports OLS on the censored durations for years to drilling. The Cox estimates are valid if censoring is independent of the event time given the bid, and censoring in this context is administrative. It depends on when the contract was signed and when the data end, not on any choice by the winner, which makes the assumption plausible, although blocks from earlier tenders are observed for longer.

OLS estimates of β are likely to be biased. The error term contains the winner's drilling cost, which I do not observe and which moves the winning bid and the later decisions together. A firm with a lower cost values the block more, so it bids a higher share and is more likely to win. The same low cost makes drilling profitable at a lower price, so the winner drills sooner. The winning bid is therefore correlated with ε_i, and OLS mixes the effect of the share on drilling with the effect of the winner's cost. Because low-cost winners both bid more and drill sooner, OLS understates the delay that a higher share causes, and it would find a higher bid going with earlier drilling even if the share itself had no effect on timing. My data do not contain enough detail on cost characteristics to control for them directly, so I use an instrumental variable to obtain variation in the winning bid that is unrelated to the block's cost.

I instrument the winning bid with the floors and ceilings the government set on the other blocks offered in the same tender, which reflects competitiveness in these blocks. Let F_j and C_j denote the minimum and maximum government share allowed on block j, and let n_t be the number of blocks offered in tender t. The instruments are the leave-one-out means

$$\bar{F}_{-i} = \frac{1}{n_t - 1} \sum_{j \in t,\, j \neq i} F_j, \qquad \bar{C}_{-i} = \frac{1}{n_t - 1} \sum_{j \in t,\, j \neq i} C_j, \tag{1}$$

computed over all 157 blocks offered, including those that were not awarded. Table A2 reports the floors and ceilings by tender. Round 1 set no ceilings, so I code its blocks at a ceiling of 100 percent.

The government may set a block's own floor and ceiling with its characteristics in mind, so the block's own values would not be valid instruments. My identifying assumption is that characteristics are independent across blocks within a tender. Under that assumption, the floors and ceilings set on other blocks are uncorrelated with ε_i. The instruments are relevant because bidders have limited capacity to bid on, and later develop, many blocks at once. High floors and ceilings on the other blocks in a tender make those blocks less attractive and push bidders toward block i. More bidders compete for block i, and its winning bid rises.

The first stage is $b_i = \pi_0 + \pi_1 \bar{F}_{-i} + \pi_2 \bar{C}_{-i} + v_i$.

Table A7 in Appendix B reports the first stage. Both instruments enter with the expected sign and are significant at the 1 percent level. A one-percentage-point increase in the average floor on the other blocks raises the winning bid by 0.89 percentage points, and the same increase in the average ceiling raises it by 0.31 percentage points. The effects are large relative to the spread of bids. A one-standard-deviation increase in the average rival floor (8.6 points) raises the winning bid by 7.7 points, about a third of the bid's standard deviation of 22.4 points. A one-standard-deviation increase in the average rival ceiling (32.3 points) raises it by 10.1 points. The instruments explain 39 percent of the variation in winning bids. The joint F-statistic is 61.9, well above the Stock and Yogo (2002) critical value of 19.93 for a maximal size distortion of 10 percent with one endogenous regressor and two instruments.

I estimate the equation for y_i by two-stage least squares for the linear outcomes. For the Cox models I use a control function, adding the first-stage residual $\hat{v}_i$ to the hazard, and compute standard errors from 400 bootstrap replications. All other standard errors are heteroskedasticity robust. Table 2 reports OLS estimates in Panel A and IV estimates in Panel B.

3.2 Results

I describe magnitudes for a 10-percentage-point increase in the winning bid, about 0.45 standard deviations. Table 2 shows that winners with higher bids drill later and for fewer years. In the IV Cox model of column (1), a 10-point higher bid lowers the hazard of starting drilling by 30 percent (a hazard ratio of 0.70), compared with 20 percent in the OLS estimate. Column (2) puts the same result in years. The IV estimate implies a delay of 1.2 years, about 30 percent of the sample mean of 3.9 years. The number of years with an exploration plan falls by 0.57 (column 4), 13 percent of the mean of 4.5. In all three columns the IV estimate is larger in absolute value than the OLS estimate, as Section 3.1 predicts if blocks that are cheaper to drill attract higher bids and are also drilled sooner.

I find no detectable effect on the intensive margins. The IV coefficient on log exploration investment in column (3) is 0.002, which implies a change of 2 percent for a 10-point higher bid. The 95 percent confidence interval runs from about -14 to $+22$ percent, so I cannot rule out moderate effects in either direction. The coefficient on wells drilled (column 5) is also small and insignificant, at 0.6 wells against a mean of 6.3.

The production results do not support the hypothesis. In the IV estimates, a 10-point higher bid raises the probability of having produced by 5 percentage points (column 6) and the hazard of first production by 11 percent (column 7), and neither effect is significant. The OLS estimates point the same way and are significant at the 1 percent level, with a higher bid raising the probability of production by 8 percentage points and the hazard of first production by 26 percent.

Column (8) reports the government's total revenue from each contract through 2024. For licences this is the base and additional royalties, and for production-sharing contracts it is the state's share of output, paid in kind and valued at the contract prices. Blocks that never produced paid nothing and enter with zero revenue. The 11 blocks that produced but have no record in the fund's data are excluded, which leaves 93. A 10-point higher bid raises revenue by $45 million in the IV estimate, significant at the 5 percent level, and by $22 million in OLS, against a mean of $28 million.

Table A9 in the appendix reports the estimates with controls for location and block area. The results are similar to the baseline for most outcomes, although the first stage is weaker, with a joint F-statistic of 13.0.

Table 2—Winning Bid and Post-Auction Outcomes

	Exploration					Production		Revenue
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Years to drilling	Years to drilling	log Expl. investment	Expl. plan-years	Wells drilled	Produced	Months to production	Government
Unit	*years*	*years*	*US$m*	*count*	*count*	*binary*	*months*	*US$m*
Model	*Cox PH*	*Linear*	*Linear*	*Linear*	*Linear*	*Linear*	*Cox PH*	*Linear*
Panel A. No instrument								
Winning bid (%)	-0.022***	0.088***	0.005	-0.046***	0.094	0.008***	0.023***	2.182*
	(0.005)	(0.016)	(0.006)	(0.013)	(0.063)	(0.002)	(0.007)	(1.180)
Panel B. Winning bid instrumented by the rival-auction mean floor and mean cap								
Winning bid (%)	-0.036***	0.119***	0.002	-0.057***	0.059	0.005	0.010	4.508**
	(0.013)	(0.023)	(0.009)	(0.018)	(0.087)	(0.003)	(0.014)	(2.175)
First-stage F	61.9	61.9	61.9	61.9	61.9	61.9	61.9	61.9
Total obs.	104	104	76	104	104	104	104	93
Uncensored obs.	76	–	–	–	–	–	38	–
Mean of dep. variable	1.382	3.913	4.127	4.510	6.308	0.365	26.839	28.093

Notes: The sample is all awarded blocks. No controls are included. Government revenue is the payments recorded by the Fondo Mexicano del Petróleo through 2024, in millions of nominal US dollars. It is zero for blocks that never produced, and blocks that produced without a record are excluded. Column (3) covers only blocks with positive exploration investment. Panel B instruments the winning bid with the mean floor and ceiling on the other blocks of the same tender, with Round 1 blocks at a ceiling of 100 percent. Panel B uses 2SLS, or a control function with 400 bootstrap replications in the Cox columns. Other standard errors are heteroskedasticity robust. Columns (1) and (7) are Cox models of right-censored durations with coefficients in log hazard ratios, so a positive coefficient means a shorter time, and their means are over uncensored blocks. Column (2) is OLS on the censored durations.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

4 A Model of the Auction and Drilling

The estimates in Table 2 measure the total effect of a higher winning bid, and that total combines two channels that work in opposite directions. A higher share lowers the return the winner keeps on each barrel, which delays drilling. A higher share is also offered by a cheaper firm, which drills sooner. The instruments do not separate the two. They shift the

winning bid by drawing more bidders to the block, and a block with more bidders is won by a cheaper firm on average, so the rival floors and ceilings change who wins as well as what the winner pays. The IV estimate is therefore the effect of a competition-induced rise in the winning bid. If cheaper winners drill sooner, it understates the delay caused by the share alone. To separate the two channels, this section sets out a model of a block, from the auction to the payoffs, in the order in which events occur. Registered firms decide whether to bid, entrants submit a government share, the winner decides when to drill, and the firm and the government divide what the block produces.

4.1 Timeline

The government leases a block to a firm that will drill and develop it. The block holds an amount of oil that is the same whoever wins it and that every registered firm knows. Firms differ in what it would cost them to drill and develop the block, and each firm knows only its own cost. Registered firms learn their costs and decide whether to pay an entry cost to bid. Each entrant then bids the share of revenue it will pay to the government, and the government awards the block to the highest bid, breaking ties at the ceiling by a cash payment. The winner gets a contract that lets it drill the block for a fixed number of months. Each month it observes the oil price, which it takes as given, and decides whether to drill now or wait. The winning share is levied on the winner's revenue but not on its cost, so a higher share lowers the return from drilling and makes waiting more attractive. Once the winner drills, it produces oil for the rest of the contract, and the government takes its share of the revenue from that production.

4.2 Primitives

Time is measured in months. I index contract months by $t = 1, 2, \ldots$, the exploration period of block ℓ lasts T_ℓ months, and payoffs are discounted at $\delta < 1$ a month. Block ℓ has N_ℓ potential entrants, the firms registered for its tender. If it is drilled, the block yields Q_ℓ barrels of oil in expectation, net of the probability of a dry hole and discounted over its production profile, so that $Q_\ell = \sum_{\tau \geq 0} \delta^\tau q_{\ell\tau}$ for a monthly output profile $q_{\ell\tau}$. The oil is a property of the block. It is the same whoever wins, and every registered firm knows it, because all of them work from the same seismic data and resource estimates in CNH's data room.

Firms differ instead in what it would cost them to drill and develop the block. Firm i's cost on block ℓ is $C_{i\ell}$ dollars. It depends on the firm's technical know-how, on the infrastructure it already operates nearby, and on its cost of capital, and only the firm knows it. The cost is the only source of heterogeneity across firms, so I define the type of firm i on

block ℓ as the block's oil per dollar of the firm's own cost,

$$x_{i\ell} \equiv \frac{Q_\ell}{C_{i\ell}} \qquad \text{(barrels per dollar)}. \tag{2}$$

Assumption 1 (Types)**.** The barrels per dollar of cost $Q_\ell/C_{i\ell}$ are independent across firms and blocks and lognormal, so that $\ln x_{i\ell} \sim N(m_{s(\ell)}, s^2)$, where $s(\ell) \in \{\text{onshore}, \text{deep water}\}$ is the block's stratum. Each firm observes Q_ℓ and its own cost before the auction but not its rivals' costs. The distribution differs only by stratum.

I write $F_\ell(x) \equiv \Phi\Big((\ln x - m_{s(\ell)})/s\Big)$ and f_ℓ for the distribution and density of the type on block ℓ. Two things follow from the assumption. First, values are private. The only component of value common to all firms, the oil, is known to all of them, so learning a rival's cost would not change what the block is worth to a firm, and there is no winner's curse. Second, cost is proportional to oil within a stratum, so in distribution a block with twice the oil costs every firm twice as much. This lets blocks of different size share one type distribution. If larger blocks were cheaper per barrel, $\ln Q_\ell$ would enter the mean of the type.

Treating the oil as common and known may be a strong assumption. If firms held private estimates of the oil, values would be interdependent. The highest bidder would tend to be the firm that most overestimated the oil, and a rational bidder would shade its bid to account for it. The model rules this out. The only uncertainty about the oil is news that arrives after the auction, namely the movements of the oil price in (3), which no bidder has when it bids.

The value of the oil moves with its price. Let P_t be the real oil price at the end of month t and $p_t = \ln P_t$, and let $\bar{P}_\ell$ denote the price in the first month of the contract, the price at which firms bid. The log price follows a random walk,

$$p_{t+1} = p_t + \mu + \sigma\varepsilon_{t+1}, \qquad \varepsilon_{t+1} \overset{\text{iid}}{\sim} N(0,1), \tag{3}$$

independent of the oil and the costs, with the volatility σ measured per month. Every firm faces the same price and takes it as given, because the price of Mexican crude is set in the world market, which OPEC dominates. I assume $\mu = -\sigma^2/2$ (Section 5.1), which keeps the expected price level constant, $\mathbb{E}[P_{t+\tau} \mid P_t] = P_t$ for all τ. If drilling starts in month t, the expected value of the oil is therefore $P_t Q_\ell$ (Appendix L.1).

4.3 When to Start Drilling

Let us denote the winner of block ℓ by W_ℓ. After the auction the oil, the winner's cost and its share are fixed. The state of the winner's problem is the revenue, in dollars, that it would

earn if it drilled in month t,

$$S_{\ell t} \equiv (1 - r_{W\ell})\, P_t\, Q_\ell, \tag{4}$$

the discounted barrels times the price times the fraction of their value that the winner keeps. The share and the price enter revenue in the same way, as factors. A share of 30 percent is equivalent, for the winner, to a permanent 30 percent fall in the oil price. Everything in (4) except the price is fixed once the auction is over, so revenue moves only with the price and in proportion to it, $S_{\ell,t+1} = S_{\ell t} R_{t+1}$, where $R_{t+1} \equiv P_{t+1}/P_t = e^{\mu+\sigma\varepsilon_{t+1}}$ is the gross return on the price. By (3) with $\mu = -\sigma^2/2$, the returns are independent over time and $E[R_{t+1}] = 1$, so revenue is expected to stay where it is, and one state variable suffices.

If the winner drills in month t it receives the revenue and pays the cost, $S_{\ell t} - C_{W\ell}$. If it waits it receives nothing now and faces the same choice next month with revenue $S_{\ell t} R_{t+1}$ and one month less. With n months left, the value of the right to drill, in dollars, is therefore

$$V_n(P_t; r, Q_\ell, C_{W\ell}) = \max\Big\{(1-r)P_t Q_\ell - C_{W\ell},\ \delta\, E_t\Big[V_{n-1}(P_{t+1}; r, Q_\ell, C_{W\ell})\Big]\Big\}, \qquad V_0 = 0, \tag{5}$$

where $r = r_{W\ell}$ is the winning share. The right expires at the end of the exploration period, which is why $V_0 = 0$. In the last month there is nothing to wait for, and $V_1(P_t; r, Q_\ell, C_{W\ell}) = \max\{(1-r)P_t Q_\ell - C_{W\ell}, 0\}$ is the static rule of Section 4.2, under which the winner drills if revenue covers cost. In general $V_n \geq \max\{S_{\ell t} - C_{W\ell}, 0\}$, and the difference is the option value of waiting. It is positive at every revenue when $n \geq 2$. A block that loses money at today's price is still worth something, because the price may rise before the deadline. And V_n is nondecreasing in n, because more time is more option value.

The price, the share and the oil enter V_n only through the revenue $S_{\ell t}$, and V_n is homogeneous of degree one in revenue and cost. Doubling both doubles the payoff from drilling in every month. The value therefore reduces to that of a winner whose cost is one dollar,

$$V_n(P_t; r, Q_\ell, C) = C\, v_n\Big((1-r)P_t Q_\ell / C\Big), \qquad v_n(S) = \max\Big\{S-1,\ \delta\, E\, v_{n-1}(SR)\Big\}, \qquad v_0 = 0, \tag{6}$$

where S is now revenue per dollar of cost and R the gross return on the price.

Proposition 1 (Trigger rule)**.** *For each $n \geq 1$ there is a threshold $S_n^* = C_{W\ell}\, e^{y_n^*}$ such that a winner with n months left drills if and only if its revenue $S \geq S_n^*$, equivalently $\ln(S/C_{W\ell}) \geq y_n^*$, with $y_1^* = 0$ and y_n^* nonnegative and nondecreasing in n. The winner drills in contract month t when*

$$P_t \ \geq\ \underbrace{\frac{C_{W\ell}}{Q_\ell}}_{\text{winner's cost per barrel}} \times \underbrace{e^{y^*_{T_\ell - t+1}}}_{\text{option value}} \times \underbrace{\frac{1}{1 - r_{W\ell}}}_{\text{bid wedge}}. \tag{7}$$

The proof is in Appendix L. In terms of revenue, the winner drills once $S_{\ell t}$ reaches

$S^*_{T_\ell - t+1}$, and the price condition (7) restates this by substituting (4). The rule separates three forces. The first factor is the cost per barrel of the firm that won, so the price at which a block is drilled depends on who won it. The second, $e^{y_n^*} = S_n^*/C_{W\ell}$, is the efficient delay, how far revenue must exceed cost before an owner paying no share would drill under price uncertainty. The third is the distortion from the share, which lowers revenue at every price by the factor $1 - r_{W\ell}$ and so raises the required price by $1/(1 - r_{W\ell})$ on top. If the price path crosses the higher trigger before T_ℓ, the share delays drilling. If it never does, the share prevents it.

4.4 Bidding

Firms bid at the contract-start price $\bar{P}_\ell$. Let $G_\ell(r)$ be the probability that every rival's share is below r, counting rivals that do not enter as bidding nothing, and g_ℓ its density. An entrant of type x, whose cost is $C = Q_\ell/x$, chooses $r \in [r_{0\ell}, \bar{r}_\ell]$, between the floor and the ceiling, to maximize $G_\ell(r)\, V_{T_\ell}(\bar{P}_\ell; r, Q_\ell, C)$, the probability of winning times the value of the right to drill with starting revenue $(1 - r)\bar{P}_\ell Q_\ell$ and T_ℓ months left. The starting revenue is the same for every bidder at a given share, because they share the oil. What differs is the cost. A higher share raises the chance of winning and lowers the revenue from winning.

Proposition 2 (Bidding first-order condition)**.** *Suppose an entrant of type x, with cost $C = Q_\ell/x$, bids an interior share $r \in (r_{0\ell}, \bar{r}_\ell)$ and V_{T_ℓ} is differentiable in r there. Then*

$$\underbrace{g_\ell(r)\, V_{T_\ell}(\bar{P}_\ell; r, Q_\ell, C)}_{\textit{competition effect}} \;=\; \underbrace{G_\ell(r)\Big(-\frac{\partial V_{T_\ell}(\bar{P}_\ell; r, Q_\ell, C)}{\partial r}\Big)}_{\textit{incentive effect}}. \tag{8}$$

The proof is in Appendix L. The left side is the competition effect. A higher share raises the probability of winning at the rate $g_\ell(r)$, and each additional win is worth the value of the right to drill. The right side is the incentive effect. A higher share lowers the revenue the winner keeps at the rate $\bar{P}_\ell Q_\ell$, and with it the value of the right in every event in which the firm wins, which has probability $G_\ell(r)$. The derivative $\partial V_{T_\ell}/\partial r$ already accounts for the later drilling that the lower revenue causes.

Let $\rho_\ell(x)$ denote the equilibrium bidding strategy on block ℓ, the share that a bidder of type x bids in a symmetric equilibrium. The share is a decreasing function of the cost, so ρ_ℓ is increasing in the type. A rival bids below $\rho_\ell(x)$ if it does not enter or its type is below x, so $G_\ell(\rho_\ell(x)) = F_\ell(x)^{N_\ell - 1}$. Substituting into (8) gives

$$\rho_\ell'(x) = (N_\ell - 1)\,\frac{f_\ell(x)}{F_\ell(x)}\,\frac{V_{T_\ell}\big(\bar{P}_\ell; \rho_\ell(x), Q_\ell, Q_\ell/x\big)}{-\partial V_{T_\ell}\big(\bar{P}_\ell; \rho_\ell(x), Q_\ell, Q_\ell/x\big)/\partial r}, \qquad \rho_\ell(x_\ell^e) = r_{0\ell}. \tag{9}$$

The bid function starts at the floor at the entry cutoff x_ℓ^e (Section 4.5) and rises until it

reaches the ceiling at a type x_ℓ^c.

Types above x_ℓ^c all offer the ceiling $\bar{r}_\ell$ and compete in cash. Among them this is a first-price auction for the right to drill with starting revenue $(1-\bar{r}_\ell)\bar{P}_\ell Q_\ell$, and the cash bid b of a firm of type x satisfies

$$V_{T_\ell}\Big(\bar{P}_\ell; \bar{r}_\ell, Q_\ell, Q_\ell/x\Big) = b + \frac{H_\ell(b)}{h_\ell(b)}, \tag{10}$$

where H_ℓ is the distribution of the highest rival cash bid and h_ℓ its density. All firms at the ceiling compete for the same revenue, so the prize differs across them only through the cost. By (6) its elasticity with respect to the cost is $1 - Sv'_{T_\ell}(S)/v_{T_\ell}(S)$ at revenue per dollar of cost S, which is negative because v_{T_ℓ} is convex and vanishes at zero, so that $Sv'_{T_\ell} > v_{T_\ell}$. A cheaper firm therefore values the same revenue more. Cash bids therefore fall with the cost, and among the firms at the ceiling the block goes to the cheapest. The ceiling type x_ℓ^c is indifferent between the highest interior share and the ceiling with a cash bid of zero.

Proposition 3 (Ranking of winners)**.** *Along an increasing equilibrium, the starting revenue per dollar of cost* $a(x) \equiv \big(1-\rho_\ell(x)\big)\bar{P}_\ell\, x$ *satisfies* $\operatorname{sign}(a'(x)) = \operatorname{sign}\left(\frac{d}{dr}\, G_\ell(r)/[(1-r)\, g_\ell(r)]\right)$ *at* $r = \rho_\ell(x)$*. If* G_ℓ *is log-concave, higher types keep a higher revenue per dollar of cost after paying their share.*

Proposition 3 describes how the auction screens firms. Under independent private costs, the auction awards the block below the ceiling to the cheapest entrant, and the cash tiebreak does the same at the ceiling. Because the oil is the same for every firm, the cheapest entrant is also the one whose drilling creates the most surplus, so among the firms that enter the government allocates the block efficiently. A share bid is a payment contingent on the outcome, however, and one may worry that the firm promising the government the most keeps the least reason to drill. The winner does bid the highest share, so its starting revenue $(1-r)\bar{P}_\ell Q_\ell$ is the lowest among the entrants. What sets the timing of drilling is revenue relative to the winner's own cost, and Proposition 3 says that this ratio rises with the type when G_ℓ is log-concave. The firm the auction selects is then the entrant that would drill first. The share still delays drilling relative to an owner paying no share, through the wedge in (7), but the delay is a distortion from taxing revenue rather than from selecting the wrong firm. Selection works against it instead. A higher winning bid induced by competition comes with a cheaper winner, who drills sooner, which is why the instrumental-variable estimates of Section 3 understate the delay caused by the share alone.

4.5 Equilibrium

Firm i knows its cost, and so its type, and decides whether to prepare a bid, which costs c_ℓ. A firm that does not bid gets zero.

Assumption 2 (Entry)**.** Entry decisions are simultaneous and made block by block, and entrants do not observe the number of other entrants when they bid. The entry cost varies across blocks so that the probability that a registered firm bids, q_s, is the same for every block in a stratum.

A cheaper firm values any given revenue more and, bidding a higher share, wins more often, so the value of bidding falls with the cost. A firm therefore enters if and only if its type exceeds a cutoff x^e_ℓ, that is, if its cost is below $C^e_\ell = Q_\ell / x^e_\ell$. Each registered firm enters with probability $q_s = 1 - F_\ell(x^e_\ell)$, so $x^e_\ell = \exp\left(m_s + s\,\Phi^{-1}(1-q_s)\right)$, and the number of bidders is binomial with parameters N_ℓ and q_s. The marginal entrant bids the floor and wins only if no rival enters, and its expected profit equals the entry cost,

$$F_\ell(x^e_\ell)^{N_\ell - 1}\; V_{T_\ell}\left(\bar{P}_\ell; r_{0\ell}, Q_\ell, C^e_\ell\right) = c_\ell. \tag{11}$$

Given the cutoff, (11) gives the entry cost of each block.

An equilibrium for block ℓ consists of the value functions V_n and thresholds y^*_n solving (5), a bid function ρ_ℓ solving (9) below the ceiling and cash bids solving (10) at it, and an entry cutoff x^e_ℓ, with the entry cost solving (11). It is computed in the order of this section. By homogeneity the exploration stage reduces to the value of a winner whose cost is one dollar, v_n in (6), and its thresholds y^*_n, which do not depend on any block-specific object and are computed once. Given v_{T_ℓ} and the cutoff, the bidding stage is an ordinary differential equation from the floor.

5 Estimation

I estimate the model by maximum likelihood on the exploration-and-extraction licences. The estimation follows the backward order of the model. The price process and the discount factor are set first, outside the likelihood, and the entry probabilities are estimated from counts of bids. The type distribution, its mean by stratum and its dispersion, and the three parameters of the exploration stage are then estimated jointly from bids and the timing of drilling.

5.1 The Oil Price Process

The value of waiting depends on how the oil price is expected to move, so I first estimate the price process (3) outside the model. Under (3), the volatility σ governs the value of waiting. A more volatile price makes it more likely that a block worth little today will be worth drilling later, so the winner requires a higher price before it commits. The price level grows in expectation by e^g with $g = \mu + \sigma^2/2$. The exploration stage needs the discounted expected

growth to be below one, $\delta e^g < 1$, because otherwise waiting always dominates drilling and no winner would drill before its deadline.

I fix the drift at $\mu = -\sigma^2/2$, which makes the expected price level constant. Then $g = 0$, so that $E[P_{t+1}] = P_t$ (Appendix L.1 derives this). A firm's revenue is proportional to the price level, so this is the natural assumption that the winner expects no trend in the price it will receive. It is also the no-change forecast, which is hard to beat for oil (Alquist et al., 2013). With $g = 0$ the condition $\delta e^g < 1$ holds for any discount factor below one, and waiting is costly only because future revenue is discounted.

I do not estimate the drift, for two reasons. Over the sample I use, the estimated annual drift is -0.096 with a standard error of 0.11, so its 95 percent confidence interval runs from -0.32 to 0.13. And the model is sensitive to it, since a small change in μ moves δe^g across one at standard discount rates. Setting $\mu = 0$ instead, a random walk without drift in the log price, is not neutral either. It implies that the expected price level rises by $\sigma^2/2$, about 3 percent a year in real terms, purely because the price is volatile.

I use daily oil prices from the CNH's Hydrocarbon Information System (SIH), take the price on the last trading day of each month, and deflate it with the US consumer price index. I restrict the sample to January 2010 through January 2015, the five years before the first auction in July 2015. This is the price history bidders could observe when they bid, so it is the process they could have estimated when they valued the right to drill. The timing of drilling is evaluated separately, on the price path that was actually realized after each contract started. Figure A2 in Appendix A plots the series. I use end-of-month prices rather than monthly averages because averaging a random walk smooths it and understates its volatility.

The model moves in monthly steps, but what matters for the value of waiting is how far the price moves over the several years of an exploration period. I therefore estimate σ from 12-month log changes, taken over every month of the sample, as the maximum-likelihood volatility around the imposed mean $-\sigma^2/2$. Because the mean depends on σ, I solve for it by iterating to a fixed point. Under an exact random walk the same estimate from monthly changes, scaled by $\sqrt{12}$, would coincide. To check the random-walk assumption I also regress the monthly log price on its own lag.

Table A5 in Appendix A reports the estimates. The annual volatility is 0.25, so a typical change in the real oil price over a year is about 25 percent. The implied drift of the log price is -0.030, which offsets the convexity term exactly and keeps the expected price level flat. Scaling the volatility of monthly changes by $\sqrt{12}$ gives almost the same figure, 0.25, and the variance of a year's change is 96 percent of what monthly changes imply, as a random walk predicts. In the model's monthly units the annual volatility is $\sigma = 0.25/\sqrt{12} = 0.071$, so that twelve monthly steps reproduce the variance of a year's change. The sample is short. Five years of data contain only about four independent one-year changes, so the standard error of the volatility is 0.086, about a third of the estimate.

5.2 Drilling

Given $\sigma = 0.071$, $\mu = -\sigma^2/2$ and $\delta = 0.991$ a month, I solve the exploration stage once, before the likelihood. Because V_n is homogeneous of degree one in revenue and cost (Section 4.3), I solve (6) once, for a winner whose cost is one dollar, and the solution serves every block and every winner. The right to drill expires, so the problem has a finite horizon, and I solve for the value functions by backward induction. The value with no months left is known, $v_0 = 0$, and so is the value in the last month, $v_1(S) = \max\{S - 1, 0\}$, when there is nothing left to wait for. Given v_{n-1}, (6) gives v_n directly. In the log revenue per dollar of cost, $y = \ln S$ for a one-dollar cost, the recursion I solve is

$$v_n(e^y) = \max\left\{ e^y - 1,\ \delta \sum_{k=1}^{21} \omega_k \, v_{n-1}\left(e^{\,y+\mu+\sigma\xi_k}\right)\right\}, \tag{12}$$

where ξ_k and ω_k are the nodes and weights of 21-point Gauss–Hermite quadrature for a standard normal shock, so the sum is the expectation over next month's price. Starting from the deadline and stepping back one month at a time, I obtain v_n for every number of months left up to the longest exploration period.

At each step I evaluate v_n on a grid of 1,401 values of y between -8 and 6 and interpolate linearly between grid points. Then, the value in dollars for a winner with cost C and revenue S is $C\, v_n(S/C)$. The thresholds y_n^*, the log of the revenue at which a winner with a one-dollar cost drills, are 0, 0.31, 0.45 and 0.48 with 1, 12, 60 and 120 months left, so a winner with five years left waits until its revenue is $e^{0.45} \approx 1.6$ times its cost. I compute the ratio $v_{T_\ell}(S)/[S\, v'_{T_\ell}(S)]$ by numerical differentiation. I set the exploration periods to 6 years onshore and 10 years in deep water, so $T_\ell = 72$ or 120 months, pending the contract terms.

Figure A3 in Appendix F plots the solution $v_n(S)$ with 1, 12, 60 and 120 months left. Below its threshold each curve lies above the payoff from drilling now, $S - 1$, because the winner can wait for a higher price, and the value of waiting grows with the months left. From the threshold on, the curve equals $S - 1$ and the winner drills.

5.3 The Likelihood

The likelihood is the probability of the observed data given the parameters $\theta = (m_{\text{onshore}}, m_{\text{deep}}, s, \pi_L, \sigma_e, \lambda)$. The data are the shares bid on the licence blocks and, for each winner, the year in which it started drilling, or the fact that it had not started by the end of 2024. Types are independent across firms and blocks, so the log likelihood is the sum of one contribution for each bid and one for each winner. I take as given each block's registered firms, floor, ceiling and contract-start price, the realized price path, the entry probability and the solution of the exploration stage.

I estimate the entry probability in each stratum by maximum likelihood, $\hat{q}_s =$

$\sum_{\ell \in s} n_\ell / \sum_{\ell \in s} N_\ell$, where N_ℓ counts the firms registered for the block's tender and n_ℓ its valid bids, over all offered blocks, including those that received no bid. It is 0.180 onshore and 0.047 in deep water. With $\ln x \sim N(m_s, s^2)$ (Assumption 1), the entry cutoff of Section 4.5 is $x^e_\ell = \exp\left(m_{s(\ell)} + s\,\Phi^{-1}(1-\hat{q}_s)\right)$.

At a trial value of θ, $m_{s(\ell)}$ and s give the distribution of types F_ℓ and its density f_ℓ, and with them the entry cutoff x^e_ℓ. Given F_ℓ, I solve the equilibrium bidding strategy (9) of each block by fourth-order Runge–Kutta from the floor at the cutoff, with the block's own floor, ceiling and number of registered firms, up to the type x^c_ℓ at which it reaches the ceiling. A share strictly below the ceiling corresponds to the type $x = \rho_\ell^{-1}(r)$ and contributes the density of an entrant's share, $f_\ell(x)/[\hat{q}_s\,\rho'_\ell(x)]$, and a share at the ceiling contributes the probability that an entrant's type exceeds the ceiling type, $[1 - F_\ell(x^c_\ell)]/\hat{q}_s$. A trial value whose F_ℓ implies bid functions that fit the observed shares poorly gets a low likelihood, and the bid function at the maximum is the equilibrium under the estimated type distribution.

Each interior bid could instead be inverted separately. By (8), a share $r_{i\ell}$ reveals the starting revenue per dollar of cost $S_{i\ell}$ that solves

$$\frac{v_{T_\ell}(S_{i\ell})}{S_{i\ell}\,v'_{T_\ell}(S_{i\ell})} = \frac{G_\ell(r_{i\ell})}{(1-r_{i\ell})\,g_\ell(r_{i\ell})}, \qquad x_{i\ell} = \frac{S_{i\ell}}{(1-r_{i\ell})\,\bar{P}_\ell}, \tag{13}$$

with G_ℓ and g_ℓ (the distribution and density of the rival bids) estimated from the distribution of rival bids, as in Guerre et al. (2000).

The estimation is a nested loop. Two objects are computed once, outside both loops, because they do not depend on θ. One is the entry probability $\hat{q}_s$. The other is the solution of the exploration stage in Section 5.2, that is, the thresholds y^*_n and the ratio $v_{T_\ell}(S)/[S\,v'_{T_\ell}(S)]$ that enters (9). The outer loop searches over θ by Nelder–Mead from five starting values, in terms of m_{onshore}, m_{deep}, $\ln s$ and $\operatorname{logit}\pi_L$. The inner loop evaluates the log likelihood at a trial value. It builds F_ℓ, solves the bid function of every block and adds the contribution of every bid, then adds the log of the likelihood for every winner.

Figure 2 plots the estimated bid function of the blocks of Round 2 Tender 3 against the type $X = Q/C$, in barrels per thousand dollars of cost, for each of the tender's three combinations of floor and ceiling. The estimated bid functions of the other tenders, including Round 2 Tender 2 with 9 registered firms, are shown in Appendix G.

5.4 Estimates and Fit

Table A10 in Appendix E reports the estimates. I compute standard errors from the inverse of the numerical Hessian of the log likelihood at the maximum, with the delta method for s and π_L. Onshore, the mean of the log type is -4.13 with a standard error of 0.05, which puts the median cost at about \$62 a barrel. In deep water the mean is -6.33 with a standard error of 0.05, a median cost of about \$560 a barrel. The dispersion of types is 0.30, so within

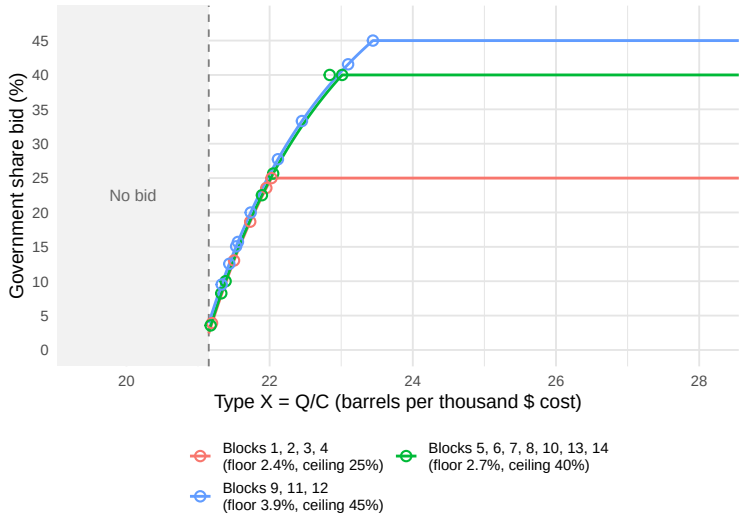


Figure 2: Estimated Bid Function, Round 2 Tender 3

Notes: Equilibrium share $\rho_\ell(X)$ at the estimates, one curve per floor and ceiling. Firms below the entry cutoff (shaded) do not bid. Circles are observed bids, placed at $\rho_\ell^{-1}(r)$ below the ceiling and at the ceiling type at the ceiling.

a stratum costs vary little, and the 90th percentile of the cost is about twice the 10th.

To check the fit, I simulate the whole market 1,000 times at the estimates. For every offered block I draw the types of the registered firms, apply the entry cutoff and the block's bid function, and pick the winner. For blocks awarded in the data, the simulated winner then decides when to start drilling on the actual price path from the actual contract start. Each statistic is computed on each simulated sample as on the data, so its spread across simulations is the sampling variation the model predicts for a sample of this size. Table 3 compares the results with the data. Entry fits well in both strata, and so do the number of bids at the ceiling and the mean winning share onshore. Drilling fits well in deep water. Interior shares are too high in the model, 0.21 against 0.16 onshore, which suggests that a single normal type distribution per stratum cannot produce both the many bids at the ceiling and the low bids below it. In deep water the model puts too many winners at the ceiling, 0.49 against 0.33, and its winning shares are slightly too high, 0.18 against 0.16. Four of the 22 statistics differ from the model's mean by more than two simulation standard errors.

6 Counterfactual Allocation Mechanisms

I use the estimates to quantify how the minimum bid of the revenue-sharing auction shapes the outcome and how the auction compares with auction formats that would not distort drilling. Throughout, the outcomes are the government's revenue and whether the winner

Table 3—Model Fit: Predicted and Observed Outcomes

	Onshore		Deep water	
	Data	Model (SE)	Data	Model (SE)
Panel A. Entry and bidding, offered blocks				
Bids per offered block	2.67	2.67 (0.30)	1.33	1.35 (0.18)
Offered blocks with no bid	0.12	0.08 (0.05)	0.31	0.25 (0.07)
Bids at the ceiling	0.64	0.64 (0.06)	0.25	0.37 (0.07)
Mean interior share	0.16	0.21 (0.02)	0.12	0.13 (0.01)
SD of interior shares	0.10	0.11 (0.01)	0.06	0.06 (0.01)
Mean winning share	0.31	0.31 (0.01)	0.16	0.18 (0.01)
Winners at the ceiling	0.81	0.84 (0.07)	0.33	0.49 (0.08)
Panel B. Drilling, winners				
Winners that started drilling by 2024	0.57	0.77 (0.10)	0.11	0.16 (0.08)
Mean lag to the start of drilling (years)	2.33	1.89 (0.50)	5.00	3.55 (1.32)
Winners that started drilling by 2024, given actual winning share	0.57	0.78 (0.09)	0.11	0.16 (0.07)

Notes: Model columns are means over 1000 simulations of the market at the estimates, with standard deviations across simulations in parentheses. Drilling is simulated on the actual price path from each contract start. Shares are fractions. The last row conditions on each winner's actual share. Lags are years from the contract year to the start of drilling.

starts drilling within the exploration period. Drilling also matters because it reveals information about the block's oil, which is valuable for neighboring blocks and for future production, beyond its direct value to the winner's own production (Kellogg, 2011). For each mechanism I simulate 2,000 markets on every onshore block, with common random numbers across mechanisms. After the assignment the operator decides when to drill as in Section 4.3.

The appendix reports two further results. The revenue-sharing auction is not always efficient ex post. Appendix H presents an example in which the winner's high share keeps it from drilling, and the government could raise both its revenue and total surplus by awarding the block to the second-highest bidder. The auction also raises drilling and revenue by screening firms through their bids. The government wants to allocate the field to a low-cost firm but cannot observe costs, and the auction lets firms reveal them through their bids. Appendix I shows that, compared with assigning each block at a fixed share to a firm drawn at random from the registered firms, awarding it to the auction winner raises the share of oil on which drilling starts within the exploration period from 59 to 78 percent when the government takes no share, and it raises the most revenue the government can collect from 99 to 122 percent of the revenue of the actual auction without ceilings.

6.1 Raising the Minimum Bid

A minimum bid is a common tool for raising the winning bid, but in a revenue-sharing auction it also reaches the winner's drilling decision. The government's revenue from a block is its share of the value of the oil produced. Raising the floor raises the share, but it delays drilling and lowers the value produced, and revenue falls once the second effect outweighs the

first. The floor changes the auction through bidding and through entry. The bid function (9) starts at the floor at the entry cutoff, so raising the floor shifts the whole bid function up. The bids of the lowest entrant types rise almost one for one, and higher types' bids rise by less, because a higher share leaves the winner's option to drill further out of the money and flattens the bid function. A firm that is the only entrant pays exactly the floor, which is common onshore, where only 18 percent of registered firms bid on a given block. The floor also lowers the value of entering, because it forces even a lone bidder to promise the government more. The entry cutoff rises, only cheaper firms enter, and more blocks draw one bid or none. Conditional on an award the winner is cheaper, which pushes toward earlier drilling, but the higher share pushes the other way.

Figure 3 traces the auction without ceilings over a common floor on every block, re-solving entry and bidding while holding each block's entry cost fixed. At low floors a higher floor raises revenue while lowering drilling, because the larger take per barrel outweighs the delay. Revenue, indexed to 100 in the actual auction without ceilings, peaks at 101 with a floor of 20 percent. Above it the incentive effect dominates, fewer blocks are awarded, and revenue falls. At a floor of 80 percent, the blocks that still attract a bid hold only 4 percent of the oil. The results show that the design of the auction should account for the winner's subsequent drilling decision as well as for the auction outcome itself. A government intervention that raises the winning bid can worsen the final outcome if the higher bid distorts the winner's drilling decision.

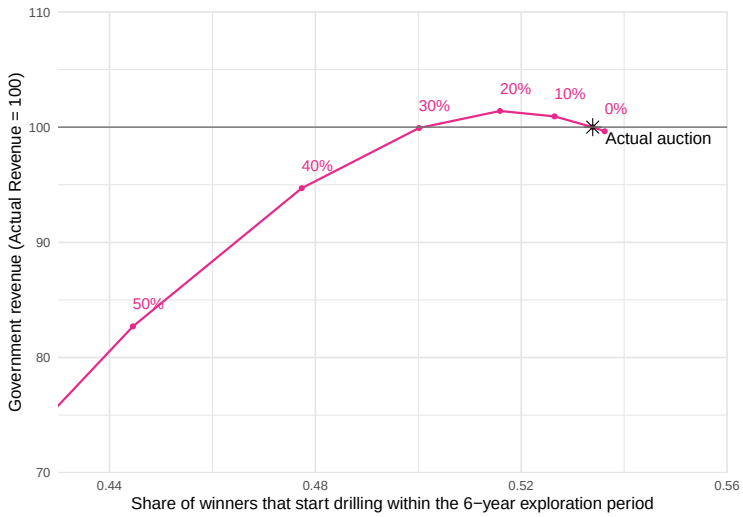


Figure 3: Revenue and Drilling in the Auction under Different Floors

Notes: Every block has a common floor (labels) and no ceiling, with entry and bidding re-solved at each block's entry cost. Floors above 50 percent fall outside the figure. Outcomes are averaged over the 24 onshore blocks, weighted by P50 prospective resources. Revenue is indexed to the actual auction without ceilings = 100.

6.2 Alternative Auctions

The actual auction has an incentive effect because the revenue share distorts the winner's drilling decision. A bonus or a share of profit has no such effect, because neither payment changes the winner's drilling decision. Table 4 compares the actual auction with a revenue-sharing auction with no floor or ceiling and with these two alternatives. In the first, firms bid a share of revenue, as in the actual auction. In the second, they bid a cash bonus paid at the start, the highest bid wins, and the winner pays its bid and no share. In the third, they bid a share of the profit, the value of the oil minus the cost of drilling and developing it. In all three the marginal entrant bids nothing and wins only if no rival enters, so the same firms enter and the same firm wins.

Onshore the revenue share delays drilling the most. Without a floor or a ceiling the winner drills within six years on 48 percent of awarded blocks, against 61 percent in the actual auction. The bonus and the share of profit leave the winner's decision to drill unchanged, because the bonus is sunk and the share of profit takes the same fraction of the payoff whenever the winner drills. Under both the winner drills on 76 percent of awarded blocks, and total surplus is the highest of the four, \$110 million per offered block against \$107 million in the actual auction. The two divide it differently. The bonus leaves most of it to the winner, and the government collects \$37 million, less than the \$53 million of the actual auction. Bidders for a share of profit compete away most of the profit, offering 79 percent of it, and the government collects \$89 million. These dollar figures scale each block's oil by the cost level recovered from the cash tiebreaks, so they are approximate, but the comparison across columns does not depend on the scale. Table A12 in Appendix J reports the same comparison for deep-water blocks, and Appendix K derives the bids in each auction.

Table 4—Outcomes of Four Auction Formats

	Actual auction	Revenue share, no floor or ceiling	Bonus, first price	Profit share
Blocks awarded (%)	91.6	92.2	92.2	92.2
Winning bid	31% of revenue	48% of revenue	\$40.2 million	79% of profit
Winner drills within 6 years (%)	61.2	47.6	75.8	75.8
Winner's revenue (million \$)	140.5	125.6	153.0	153.0
Winner's profit (million \$)	53.7	40.3	72.9	21.2
Government revenue (million \$)	53.2	59.3	37.0	88.7
Total surplus (million \$)	106.9	99.6	110.0	110.0

Notes: Simulated at the estimates, 2000 replications per offered onshore block with the same draws in every column. The actual auction uses each block's floor and ceiling, and the other columns have neither. Drilling is among awarded blocks. Money is in million US dollars per offered block, discounted to the contract start, and total surplus excludes entry costs. Each block's oil is scaled so that the median winner's cost is the \$66 million recovered from the cash tiebreaks.

Although the bonus and profit-sharing auctions raise total surplus, they are hard to implement in the countries that use revenue sharing. In a bonus auction the winner pays

a fixed sum up front, whatever the field turns out to hold, so it bears all the risk of the uncertain reserve. Where insurance and financial markets are thin, that risk deters entry and lowers bids. A profit-sharing auction requires the government to know the winner's drilling and development costs. Those costs are hard to monitor, and the winner has an incentive to overstate them, which would erode the government's share of the profit.

7 Conclusion

In a revenue-sharing auction the winning bid does not end with the auction. The winner pays its share of every barrel it later produces, so the bid acts as a tax on the return to drilling. In the model this tax raises the price at which the winner is willing to drill by the factor $1/(1-r)$, which delays drilling or prevents it before the lease expires. A government that judges the auction by the winning bid alone therefore misses part of what the bid costs it. The evidence from Mexico's oil auctions supports this concern. Winners with higher bids start drilling later, and the instrumental-variable estimates show that a 10-percentage-point higher bid lowers the hazard of starting to drill by 30 percent. The reduced form mixes this incentive effect with the selection of cheaper winners, which pushes the other way, so I estimate a model of entry, bidding and the timing of drilling to separate the two. The estimates show that the auction is valuable as a screen. Awarding a block to the auction winner rather than to a randomly chosen firm raises both drilling and the most revenue the government can collect. But the same screen ties the government's take to the winner's incentive to invest. Spreading shares across winners delays drilling more than a uniform share would, and raising the minimum bid beyond about 20 percent lowers the government's revenue as well as drilling.

Auction formats that leave the drilling decision undistorted would do better. A bonus auction or a profit-sharing auction raises the share of blocks drilled within the exploration period and the total surplus, and the two differ mainly in how they divide that surplus between the winner and the government. Neither is easy to adopt in the countries that rely on revenue sharing. A bonus asks bidders to bear the risk of an uncertain field up front where insurance and credit are scarce, and a share of profit requires the government to verify costs that the winner has every reason to overstate. Within revenue-sharing, the results caution against rules that push the winning bid up, such as high minimum bids, because the government collects its share only from the fields that are actually drilled.

The same lesson applies wherever the winner of an auction makes a later decision that the auctioneer cares about. In pay-as-produced power purchase auctions, the lowest price that competition delivers can discourage the winner from generating, just as a high revenue share discourages drilling. Two extensions would sharpen the analysis. First, the estimates capture only the extensive margin of production, whether a block is drilled at all, and not how much

it produces once drilled. A delay in drilling also delays production, so total production over a finite lease can fall even if the winner drills. My estimates of the effects of the auction rules on production and government revenue are therefore lower bounds. Second, the model treats blocks as independent, while the reduced form suggests that the floors set on one block shift competition for the others. Accounting for both would show more fully how auction rules shape what the government finally collects.

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A Additional Descriptive Statistics

Table A1—Summary Statistics

Variable	Obs.	Mean	SD	Min	Max
A. Auction					
Winning bid (government share, percent)	104	39.61	22.44	3.91	85.69
All valid bids (government share, percent)	369	33.76	21.02	2.01	85.69
Floor (percent)	104	9.54	10.15	1.00	40.00
Ceiling (percent), blocks with a ceiling	66	43.64	21.95	20.00	75.00
Mean floor on other blocks in the tender (percent)	104	8.62	8.64	2.30	34.23
Mean ceiling on other blocks in the tender (percent)	104	64.17	32.27	20.00	100.00
B. Block characteristics					
Area (km^2)	102	828.20	972.46	7.00	3,287.00
Shallow water	104	0.30	0.46	0.00	1.00
Deep water	104	0.26	0.44	0.00	1.00
Extraction contract	104	0.27	0.45	0.00	1.00
Prospective resources, P50 (million boe)	74	1,657.28	1,886.83	6.60	5,479.10
C. Post-auction outcomes					
Years to first exploration plan	104	3.91	4.22	1.00	11.00
Exploration investment (million USD)	104	74.98	87.27	0.00	448.11
Years with an exploration plan	104	4.51	3.03	0.00	9.00
Wells attributed to the contract	104	6.31	16.16	0.00	98.00
Produced by January 2025	104	0.37	0.48	0.00	1.00
Months to first production	104	68.77	35.14	7.00	120.02
Cumulative production (thousand boe)	104	1,315.23	6,213.28	0.00	51,569.33
Government revenue (million USD)	93	28.09	138.04	0.00	1,086.76
D. Oil price					
Real Brent price, end of month (USD per barrel)	122	78.13	18.89	18.40	136.87

Notes: Panels A to C describe the 104 awarded blocks, except the row for all valid bids. Bids, floors and ceilings are from CNH's result fact sheets, block characteristics from its technical sheets, exploration and production from SIH, and government revenue from the FMP through 2024. Durations are censored at the end of the data. The oil price is the end-of-month real Brent price, January 2015 to February 2025.

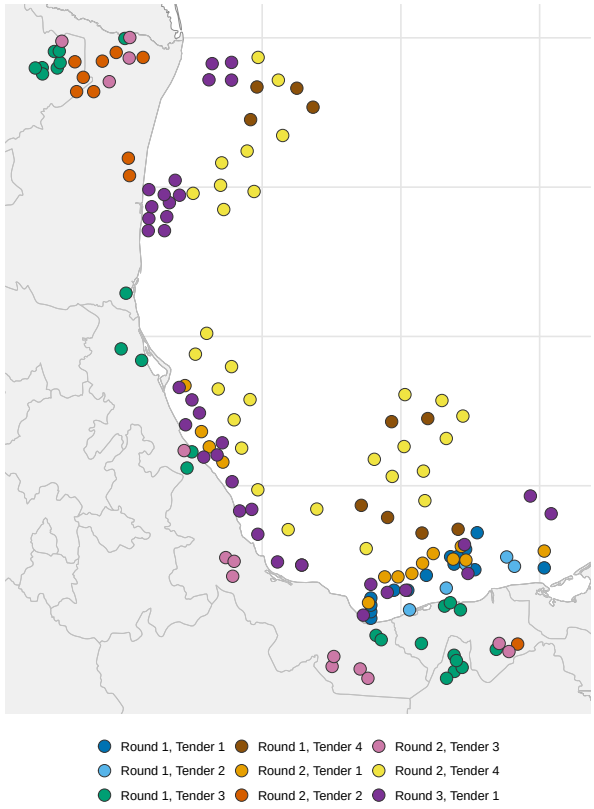


Figure A1: Location of the Blocks Offered, by Tender

Table A2—Floors and Ceilings on the Government Share, by Tender

		Floor	Ceiling	Winning bid	
Round	Tender	Mean (%)	Mean (%)	Awarded	Mean (%)
1	1	35	–	2	62
1	2	34	–	3	76
1	3	4	–	25	55
1	4	2	–	8	14
2	1	20	75	10	57
2	2	3	27	7	24
2	3	3	37	14	35
2	4	4	20	19	16
3	1	17	65	16	46
All		12	46	104	40

Notes: Each row is one tender. The floor and ceiling are the minimum and maximum government share set for each block. Round 1 set no ceilings. The mean ceiling is over capped blocks and the mean winning bid over awarded blocks.

Table A3—Number of Bidders per Block, Blocks with at Least One (Left) or at Least Two (Right) Bids

		≥ 1 bidder							≥ 2 bidders						
Round	Tender	N	Mean	P10	P25	P50	P75	P90	N	Mean	P10	P25	P50	P75	P90
1	1	6	1.8	1.0	1.0	1.0	1.8	3.5	2	3.5	2.3	2.8	3.5	4.2	4.7
1	2	3	5.0	1.8	3.0	5.0	7.0	8.2	2	7.0	5.4	6.0	7.0	8.0	8.6
1	3	25	1.0	1.0	1.0	1.0	1.0	1.0	0	–	–	–	–	–	–
1	4	8	1.6	1.0	1.0	1.0	2.0	2.6	3	2.7	2.0	2.0	2.0	3.0	3.6
2	1	10	2.8	1.0	1.0	2.0	4.8	5.1	6	4.0	2.0	2.5	4.5	5.0	5.5
2	2	7	1.7	1.0	1.0	2.0	2.0	2.4	4	2.2	2.0	2.0	2.0	2.2	2.7
2	3	14	3.7	1.3	3.0	3.0	3.8	7.4	12	4.2	3.0	3.0	3.0	4.5	7.8
2	4	19	1.7	1.0	1.5	2.0	2.0	2.0	14	2.0	2.0	2.0	2.0	2.0	2.0
3	1	16	1.6	1.0	1.0	2.0	2.0	2.0	9	2.0	2.0	2.0	2.0	2.0	2.0

Table A4—Distribution of Submitted Bids for Awarded Blocks, by Tender

		Share bid (%)							Effort bid (%)						
Round	Tender	N	Mean	P10	P25	P50	P75	P90	N	Mean	P10	P25	P50	P75	P90
1	1	7	56.27	43.63	55.27	57.00	65.06	66.66	7	21.71	5.60	8.00	10.00	17.50	46.40
1	2	15	63.01	48.86	55.80	63.84	69.12	74.66	15	13.80	0.00	0.00	0.00	11.00	33.60
1	3	25	55.27	34.95	41.50	60.74	66.30	79.93	25	73.60	7.20	30.00	100.00	100.00	100.00
1	4	13	12.85	5.33	7.44	10.39	17.01	22.21	13	0.96	0.00	1.00	1.00	1.50	1.50
2	1	28	54.96	28.87	43.08	57.91	69.74	75.00	28	0.68	0.00	0.00	1.00	1.00	1.50
2	2	12	19.08	5.63	8.68	20.98	25.00	25.00	12	1.08	0.10	1.00	1.25	1.50	1.50
2	3	52	32.39	12.58	24.64	40.00	40.00	45.00	52	1.12	0.00	1.00	1.50	1.50	1.50
2	4	33	13.95	6.13	9.03	13.44	20.00	20.00	33	0.77	0.00	0.00	1.00	1.50	1.50
3	1	25	45.43	28.50	34.73	40.51	65.00	65.00	25	0.54	0.00	0.00	0.00	1.00	1.50



Figure A2: Monthly Real Oil Price, 2010–2015

Notes: End-of-month real Brent price from SIH, in December 2024 US dollars. The window is the one used to estimate the price process.

Table A5—Real Oil Price Process, Monthly Data

Volatility σ	0.247
	(0.086)
Drift $\mu = -\sigma^2/2$	−0.030
Monthly observations	60

Notes: End-of-month real Brent price from SIH, January 2010 to January 2015, in December 2024 US dollars. The log price follows $p_{t+1} - p_t = \mu + \sigma\varepsilon_{t+1}$ with $\mu = -\sigma^2/2$, so the expected price level is constant. σ is the annual volatility of overlapping 12-month log changes. Standard error in parentheses.

Table A6—Floors, Ceilings and Block Characteristics

	Floor (%) (1)	Has a ceiling (2)	Ceiling (%) (3)
Block area (1,000 km^2)	-3.66**	-0.03	-1.40
	(1.41)	(0.12)	(1.29)
Shallow water	22.76***	-0.19**	37.43***
	(1.71)	(0.07)	(2.34)
Deep water	9.78***	-0.55**	-6.16*
	(3.26)	(0.26)	(3.29)
Extraction contract	2.33**	-0.95***	
	(1.17)	(0.04)	
Announced resources (billion boe)	0.09	0.12***	-1.28***
	(0.23)	(0.03)	(0.26)
Constant	2.48***	0.97***	34.36***
	(0.68)	(0.04)	(2.10)
R^2	0.86	0.78	0.94
Blocks	100	100	64
Mean of dep. variable	9.81	0.64	43.98

Notes: OLS on the awarded blocks. Column (3) uses only the blocks of Rounds 2 and 3, the only ones with a ceiling. Location indicators are relative to onshore. Announced resources are P50 prospective resources for exploration blocks and 2P reserves for discovered fields. Robust standard errors in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

B First Stage of the Instrumental-Variable Estimates

Table A7—First Stage: Rival Mean Floor and Ceiling, All Awarded Blocks

	Winning bid (%)
Rival mean floor (%)	0.887***
	(0.207)
Rival mean cap (%)	0.312***
	(0.060)
Constant	11.938***
	(2.586)
Joint F, rival floor and cap	61.9
R^2	0.393
Observations	104

Notes: OLS of the winning bid on the 104 awarded blocks. Rival mean floor and cap are the means over the other blocks of the same tender, with Round 1 blocks at 100 percent. This is the first stage of Panel B in Table 2. Robust standard errors in parentheses. The joint F tests that both coefficients are zero.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

C Determinants of Submitted Bids

Table A8—Determinants of Submitted Bids

	Submitted bid (%)		
	(1)	(2)	(3)
Block			
Block area (1,000 km^2)	-1.63		-0.82
	(2.49)		(1.85)
Shallow water	10.22		20.41
	(13.28)		(12.88)
Deep water	2.66		
	(6.07)		
Extraction contract	5.73		2.65
	(4.92)		(5.11)
Announced resources (billion boe)	-2.05**		-2.38**
	(0.98)		(1.06)
Floor (%)	0.69*		0.24
	(0.41)		(0.39)
Ceiling (%)	0.19		0.02
	(0.17)		(0.17)
No ceiling	-0.59		-5.28
	(5.65)		(6.00)
Bidder			
Consortium size	-0.53	-1.02	
	(1.30)	(1.35)	
Foreign member	1.28	-1.22	
	(2.65)	(2.94)	
State oil company member	-4.33	-6.89*	
	(3.08)	(4.01)	
Prior tenders entered	1.09	-0.08	
	(1.02)	(1.23)	
Holds a field in the same area	-3.04	-3.24	
	(2.43)	(2.79)	
Fields held	-0.44	-0.51*	
	(0.32)	(0.28)	
Production held (million boe/day)	50.44	60.15*	
	(37.45)	(32.76)	
Competition			
Valid bids on the block	1.02**		1.31***
	(0.41)		(0.45)
Registered bidders in the tender	-0.06		0.05
	(0.10)		(0.11)
Constant	16.41**		
	(7.35)		
Block fixed effects	No	Yes	No
Bidder fixed effects	No	No	Yes
R^2	0.47	0.58	0.74
Bids	338	310	325
Blocks	100	72	99
Mean of dep. variable	34.00	34.74	33.83

Notes: OLS of every valid bid, winning and losing. Column (2) adds block and column (3) bidder fixed effects. Location indicators are relative to onshore, and Round 1 blocks have a ceiling of zero. Holdings are what any member of the bidder held before the auction. Standard errors clustered by block in parentheses.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

D Location Fixed Effects and Block Area

Table A9—Winning Bid and Post-Auction Outcomes, Controlling for Location and Block Area

	Exploration					Production		Revenue
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Years to drilling	Years to drilling	log Expl. investment	Expl. plan-years	Wells drilled	Produced	Months to production	Government
Unit	*years*	*years*	*US$m*	*count*	*count*	*binary*	*months*	*US$m*
Model	*Cox PH*	*Linear*	*Linear*	*Linear*	*Linear*	*Linear*	*Cox PH*	*Linear*
Panel A. No instrument								
Winning bid (%)	-0.031***	0.096***	0.038***	-0.042***	-0.021	0.005**	-0.004	2.299*
	(0.008)	(0.016)	(0.008)	(0.013)	(0.076)	(0.002)	(0.010)	(1.261)
Block area (1,000 km^2)	0.464	-2.296***	0.201	1.901***	-2.051	-0.250***	-5.370***	-98.480*
	(0.312)	(0.796)	(0.264)	(0.640)	(2.127)	(0.088)	(1.608)	(52.396)
Shallow water	2.397***	-4.646***	-0.208	3.354***	-10.411***	-0.519***	-0.755	76.964
	(0.377)	(0.784)	(0.383)	(0.630)	(3.614)	(0.098)	(0.576)	(48.335)
Deep water	0.052	2.437	1.474**	-1.571	-8.708	-0.034	-10.535	263.885*
	(0.767)	(1.808)	(0.669)	(1.495)	(7.263)	(0.222)	(4,713.859)	(140.438)
Panel B. Winning bid instrumented by the rival-auction mean floor and mean cap								
Winning bid (%)	-0.221***	0.363***	0.092**	-0.209***	0.330	0.019***	0.003	11.555**
	(0.057)	(0.066)	(0.035)	(0.047)	(0.203)	(0.006)	(0.065)	(4.552)
Block area (1,000 km^2)	-0.680	0.823	0.376	-0.058	2.045	-0.082	-5.051	-10.731
	(1.074)	(1.104)	(0.308)	(0.841)	(2.314)	(0.070)	(3.252)	(32.537)
Shallow water	5.951***	-8.701***	-1.338	5.902***	-15.739***	-0.737***	-0.938	-63.960
	(1.521)	(1.864)	(0.809)	(1.257)	(5.398)	(0.124)	(2.700)	(68.848)
Deep water	-0.902	2.852	1.956**	-1.832	-8.162	-0.012	-11.048**	308.380**
	(2.343)	(2.333)	(0.843)	(1.617)	(7.448)	(0.196)	(4.756)	(131.880)
First-stage F	13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0
Total obs.	102	102	76	102	102	102	102	92
Uncensored obs.	76	–	–	–	–	–	37	–
Mean of dep. variable	1.382	3.784	4.127	4.598	5.843	0.363	27.133	28.398

Notes: The sample is all awarded blocks with a recorded area. Controls are an indicator for shallow water, an indicator for deep water (onshore is the base), block area in thousands of square kilometers. Government revenue is the payments recorded by the Fondo Mexicano del Petróleo through 2024, in millions of nominal US dollars. It is zero for blocks that never produced, and blocks that produced without a record are excluded. Column (3) covers only blocks with positive exploration investment. Panel B instruments the winning bid with the mean floor and ceiling on the other blocks of the same tender, with Round 1 blocks at a ceiling of 100 percent. Panel B uses 2SLS, or a control function with 400 bootstrap replications in the Cox columns. Other standard errors are heteroskedasticity robust. Columns (1) and (7) are Cox models of right-censored durations with coefficients in log hazard ratios, so a positive coefficient means a shorter time, and their means are over uncensored blocks. Column (2) is OLS on the censored durations.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

E Structural Estimates

Table A10—Structural Estimates: Exploration-and-Extraction Licences

	Estimate
Mean type, onshore (m_{onshore})	−4.130
	(0.049)
Mean type, deep water (m_{deep})	−6.328
	(0.052)
Dispersion of types (s)	0.299
	(0.027)
Probability of a one-year lag in the recorded start (π_L)	0.729
	(0.163)
Entry probability, onshore / deep water	0.180 / 0.047
Coefficient on the wedge (ψ), unrestricted	11.558
Likelihood-ratio statistic for $\psi = 1$	21.933
Offered blocks / winners / started by 2024	63 / 48 / 15
Log likelihood	-34.90

Notes: Joint maximum likelihood of the bids and the year in which each winner starts drilling, for the exploration-and-extraction licences of Round 1 Tender 4 and Round 2 Tenders 2–4, with $\ln(Q/C) \sim N(m, s^2)$ and a mean by location. The exploration stage uses $\sigma = 0.25$ and $\delta = 0.90$ a year. Standard errors from the inverse Hessian in parentheses. Exploration periods are provisional.

F Value of the Right to Drill

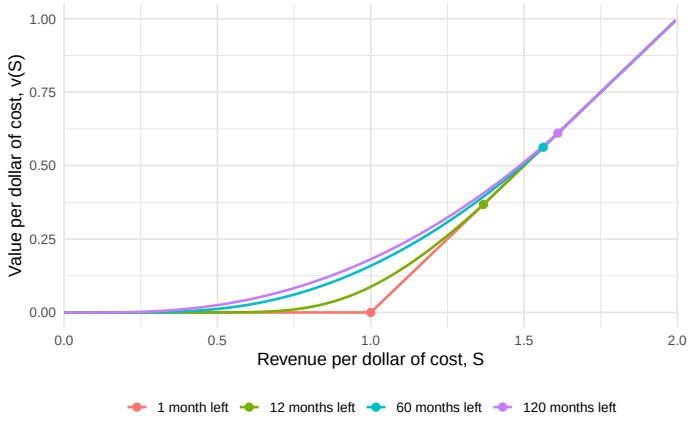
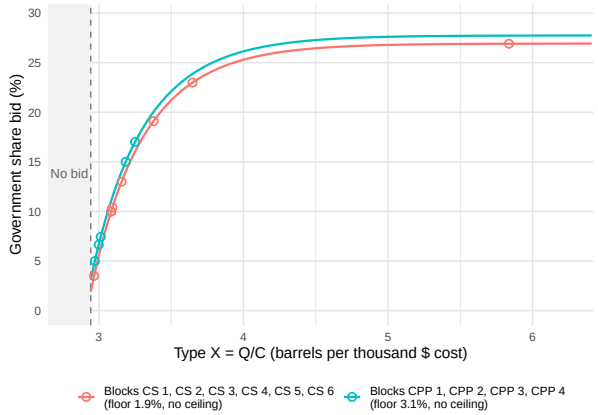


Figure A3: Value of the Right to Drill for a Winner with a One-Dollar Cost

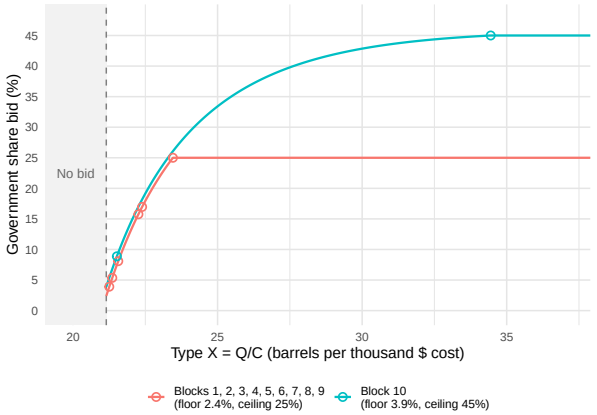
Notes: Each curve is $v_n(S)$, the value of a winner with a one-dollar cost and revenue S, with n months left. The dashed line is the payoff from drilling now, $S - 1$, and points mark the thresholds $e^{y_n^*}$ above which the winner drills.

G Estimated Bid Functions by Tender

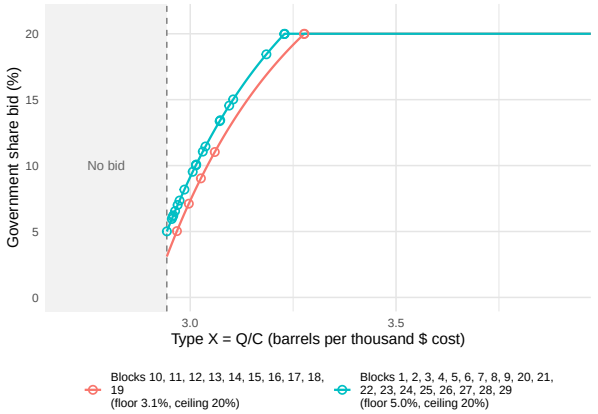
Figure A4 plots the estimated bid function of every other tender in the structural sample, drawn as in Figure 2, which shows Round 2 Tender 3.



(a) Round 1 Tender 4, Deep Water



(b) Round 2 Tender 2, Onshore



(c) Round 2 Tender 4, Deep Water

Figure A4: Estimated Bid Functions of the Other Tenders

H The Second-Highest Bidder Can Bring More Revenue

The auction awards the block to the entrant with the highest type. Because the oil is the same for every firm, this is also the entrant whose drilling creates the most surplus at any given time, so the allocation is efficient ex ante. However, the share the winner bids is a payment contingent on drilling. The cheapest firm bids the highest share, so it keeps the smallest fraction of the oil's value, and the bid wedge in (7) is largest for the firm that wins. If the wedge outweighs the winner's cost advantage, the winner drills later, or not at

all, than the second-highest bidder would have. The government would then have collected more by awarding the block to the second-highest bidder at its own lower share, and surplus would have been higher too. If this happens, the auction is not ex-post efficient, because the allocation it produces is not the one that maximizes total surplus after the auction.

I build the example from the blocks of Round 2 Tender 3 with a floor of 3.9 percent and a ceiling of 45 percent, which include Area 9 and whose bid function is drawn in Figure 2. I keep the estimated type distribution, the floor, the entry probability and the price process of Area 9, and change two things. I remove the ceiling and raise the number of registered firms from 19 to 40. I place two bidders on the resulting equilibrium bid function, a winner that bids 60 percent and a second-highest bidder that bids 35 percent. Each bidder then decides when to drill by the trigger rule of Proposition 1, which is the drilling problem that bidders anticipate when they bid. I simulate both bidders on the same 50,000 price paths over the six-year exploration period.

Table A11 reports the outcome. The winner's type is 22.8 barrels per thousand dollars of cost and the second-highest bidder's is 21.7, so the winner's cost is about 5 percent lower. The winner keeps 40 percent of the oil's value and the second-highest bidder 65 percent, so the winner's starting revenue per dollar of cost, $(1 - r)\bar{P}_\ell x$, is about a third lower. The incentive effect of the share outweighs the effect of the lower cost. The winner starts drilling within the period with probability 15 percent, against 42 percent for the second-highest bidder. The government expects $15.0 million from the winner and would have expected $19.8 million from the second-highest bidder, so the higher share does not make up for the lower chance of drilling. The second-highest bidder would also have earned more itself, and total surplus would have been $32.0 million instead of $17.8 million. The allocation the auction produces is efficient before drilling, in the sense that it selects the cheapest firm, but it is inefficient after the auction, because the share it extracts keeps that firm from drilling.

Table A11—An Example of Ex-Post Inefficiency in the Auction

	Winner	Second-highest bidder
Bid (revenue share, %)	60	35
Type Q/C (barrels per thousand $ of cost)	22.8	21.7
Drills within 6 years (%)	15.1	42.4
Bidder's expected profit (million $)	2.8	12.3
Government's expected revenue (million $)	15.0	19.8
Total surplus (million $)	17.8	32.0

Notes: Area 9 of Round 2 Tender 3 with no ceiling and 40 registered firms instead of 19. Both bids are equilibrium best responses. Values are expected present values at the contract start, with the block's oil valued at $122 million.

I The Screening Effect of the Auction

The government wants to allocate each block to the firm that can drill it most cheaply, but it cannot observe drilling costs. The revenue-sharing auction screens firms through their bids, because the cheapest entrant bids the highest share and wins. To measure this screening benefit, I compare the auction with two bilateral revenue-sharing contracts in which the government sets the share r itself. In the first, which is the worst case, the government leases the block at share r to a firm drawn at random from all registered firms, so there is no screening at all. This is a benchmark that the government can implement without knowing the firms' costs and without running any screening mechanism. In the second, which is the first best among fixed-share contracts, the government leases the block at share r to the lowest-cost firm, the firm that wins the auction. This contract is infeasible in practice, because the government cannot observe firms' costs and so cannot identify that firm without the auction, but it bounds what screening can achieve. In both contracts the share is the same for every block and every operator, and I trace each contract over shares from 0 to 99 percent.

A higher share raises what the government takes from each barrel but weakens the operator's incentive to drill. Less drilling is costly beyond the revenue it forgoes, for the reasons given at the start of Section 6.

Figure A5 plots the two contracts on the plane of drilling and government revenue, averaging over blocks with weights equal to their oil, and marks the actual auction without ceilings with a star. Raising the share first raises revenue and then lowers it, once the delay in drilling outweighs the larger take per barrel, as in a standard Laffer curve. Screening shifts the curve outward, mostly along the drilling axis. When the government takes no share of revenue ($r = 0$), a randomly chosen firm starts drilling within the period on 59 percent of the oil, and the lowest-cost firm on 78 percent. The most the government can raise is 99 percent of the revenue of the actual auction without ceilings from a randomly chosen firm, and 122 percent from the lowest-cost firm. Revenue gains less than drilling because the oil is the same whoever operates the block.

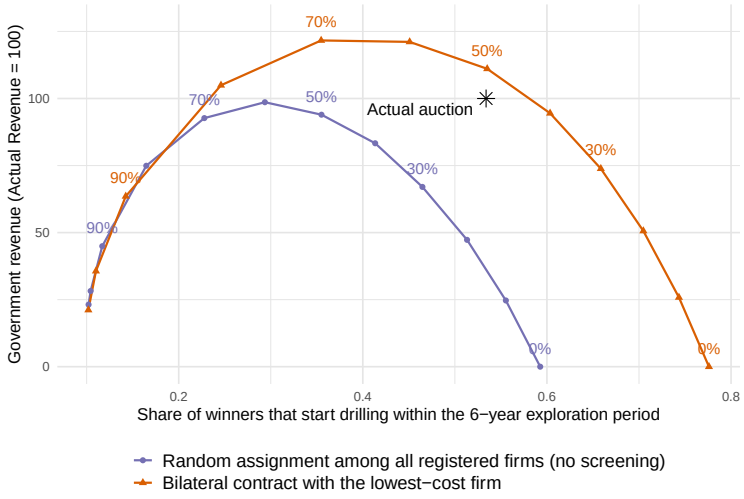


Figure A5: Screening Effect of Revenue-Sharing Auctions

Notes: Random assignment leases each block to a firm drawn from the registered firms, and the bilateral contract to the lowest-cost firm, both at share r (labels). The star marks the actual auction without ceilings. Outcomes are averaged over the 24 onshore blocks, weighted by P50 prospective resources. Revenue is indexed to the star = 100.

The actual auction lies between the two curves, close to the frontier of the lowest-cost firm but inside it. It awards the block to the same firm as that contract but charges each winner its own bid, 47 percent on average, and 53 percent of its winners start drilling within the period. Charging the same winners a uniform share of 50 percent instead gives 54 percent and a revenue of 111. The difference comes from the spread of the bids. A share r lowers the log of the winner's revenue in (4) by $-\ln(1-r)$, which is convex in r, so shares spread around the same mean delay drilling more on average than a uniform share. The winners that bid the highest shares drill latest, and the government waits longest for its revenue on exactly those blocks. This result shows that the auction succeeds in selecting the lowest-cost firm, so its outcome is close to the first best.

J Alternative Auctions for Deep-Water Blocks

Table A12—Outcomes of Four Auction Formats, Deep-Water Blocks

	Actual auction	Revenue share, no floor or ceiling	Bonus, first price	Profit share
Blocks awarded (%)	74.7	77.5	77.5	77.5
Winning bid	18% of revenue	20% of revenue	\$0.4 million	55% of profit
Winner drills within 6 years (%)	4.9	4.5	7.3	7.3
Winner's revenue (million \$)	20.2	20.1	25.6	25.6
Winner's profit (million \$)	6.9	6.6	10.9	4.7
Government revenue (million \$)	3.6	4.0	0.3	6.5
Total surplus (million \$)	10.5	10.6	11.2	11.2

Notes: Deep-water blocks, simulated and reported as in Table 4. Deep water has no cash tiebreaks, so its costs are the onshore level scaled by the ratio of planned exploration investment (\$133 against \$37 million).

K Bids in the Counterfactual Auctions

This appendix describes how I compute the bids and outcomes of the auctions compared in Table 4. Each of them sells block ℓ to one of its N_ℓ registered firms, whose types x are drawn from the estimated distribution F_ℓ, and none of them has a floor or a ceiling.

K.1 The Value of Winning and Entry

With no share of revenue, the winner of type x keeps the whole value of the oil and bears the whole cost $C = Q_\ell/x$. Its value of winning is the value of the right to drill at the full revenue $\bar{P}_\ell Q_\ell$,

$$w_\ell(x) \equiv V_{T_\ell}\Big(\bar{P}_\ell; 0, Q_\ell, Q_\ell/x\Big) = \frac{Q_\ell}{x}\, v_{T_\ell}\Big(\bar{P}_\ell\, x\Big), \tag{14}$$

where the second equality uses the homogeneity of the value function. The value rises with the type, because a cheaper firm values the same revenue more. As in the estimation, firms value the block without anticipating the news that arrives after the auction.

In all three auctions the firm with the lowest type that enters bids nothing. It wins only if no rival enters, and then it keeps the whole value and pays nothing. A firm enters if and only if its type exceeds the cutoff x_ℓ^e that solves

$$F_\ell(x_\ell^e)^{N_\ell - 1}\, w_\ell(x_\ell^e) = c_\ell, \tag{15}$$

where c_ℓ is the entry cost of the block recovered from (11) at the estimates and held fixed. The cutoff is the same in the three auctions, so the same firms enter and, because every bid rises with the type, the same firm wins. A rival bids below a firm of type x if it does not

enter or its type is below x, which has probability $F_\ell(x)$, so a firm of type x that bids as in equilibrium wins with probability $G_\ell(x) \equiv F_\ell(x)^{N_\ell - 1}$.

K.2 The Bonus Auction

In the bonus auction each entrant bids a cash payment, and the highest bid wins and pays its bid. The winner then pays no share, so its value of winning is $w_\ell(x)$. In a symmetric increasing equilibrium $b_\ell(x)$, the expected payoff of an entrant of type x is $u_\ell(x) = G_\ell(x)\,[w_\ell(x) - b_\ell(x)]$. By the envelope theorem $u_\ell'(x) = G_\ell(x)\,w_\ell'(x)$, and the marginal entrant has $u_\ell(x_\ell^e) = G_\ell(x_\ell^e)\,w_\ell(x_\ell^e)$ because it bids nothing. Integrating and solving for the bid gives

$$b_\ell(x) = w_\ell(x) - \frac{G_\ell(x_\ell^e)\,w_\ell(x_\ell^e) + \int_{x_\ell^e}^{x} G_\ell(t)\,w_\ell'(t)\,dt}{G_\ell(x)}, \tag{16}$$

which equals zero at the cutoff and lies below the value above it. I evaluate (16) on a grid of types and compute the integral by the trapezoidal rule. The bonus is paid at the contract start, so it does not enter the winner's decision to drill, which is that of a winner with no share.

K.3 The Profit-Sharing Auction

In the profit-sharing auction each entrant bids a share r of the profit, the value of the oil minus the cost, that it pays to the government when it drills, and the highest share wins. A winner that drills in month t keeps $(1-r)(P_tQ_\ell - C)$, a fixed fraction of what it would keep with no share. Its problem in the drilling stage is therefore the problem of a winner with no share multiplied by $1-r$. The share of profit does not change when it drills, and its value of winning is $(1-r)\,w_\ell(x)$.

An entrant of type x that bids as type z expects $G_\ell(z)\,[1 - r_\ell(z)]\,w_\ell(x)$. Setting the derivative with respect to z to zero at $z = x$ gives

$$r_\ell'(x) = (N_\ell - 1)\,\frac{f_\ell(x)}{F_\ell(x)}\,\big[1 - r_\ell(x)\big], \qquad r_\ell(x_\ell^e) = 0, \tag{17}$$

in which the value $w_\ell(x)$ cancels. The solution is

$$1 - r_\ell(x) = \left[\frac{F_\ell(x_\ell^e)}{F_\ell(x)}\right]^{N_\ell - 1}. \tag{18}$$

The share a firm bids depends only on the chance that every rival is below it, because both its gain from winning and its payment are proportional to its own value. The government

collects the share r of the profit realized when the winner drills. A winner drills only when the value of the oil covers the cost, so this payment is never negative.

K.4 The Revenue-Sharing Auction

The revenue-sharing auction is the model of Section 4.5 with a floor of zero and no ceiling. The bid function solves (9) from $\rho_\ell(x_\ell^e) = 0$ at the cutoff (15), and the winner pays its share of the value of the oil. Unlike the bonus and the share of profit, the share of revenue lowers the revenue the winner keeps and raises the price at which it drills, as in (7).

K.5 Outcomes

For every offered block I simulate 2,000 markets, with the same draws of types, news, prices and exogenous events in every auction. Given the winner and the share it pays, the winner drills in the first month in which its revenue per dollar of cost reaches the threshold of Proposition 1. When the exogenous event arrives it drills only if the value of the oil covers the cost, so that no winner drills at a loss. Revenue, profit and surplus are discounted to the contract start.

The model identifies costs per barrel but not the dollar size of a block. To report money in dollars I scale the oil of each onshore block so that the median winner of the actual auction has the median cost recovered from the cash tiebreaks, \$66 million. No deep-water block had a cash tiebreak. For deep water I multiply that cost by the ratio of the median planned exploration investment of deep-water winners to that of onshore winners, \$133 million against \$37 million, which gives a median deep-water winner's cost of \$241 million. The scale is common to all columns of a panel, so it does not affect the comparison across auctions.

L Proofs

L.1 Expected Price Level under the Price Process

This subsection shows that fixing the drift at $\mu = -\sigma^2/2$ in (3) keeps the expected price level constant, and what that implies for the log price.

Expected price level. Let $P_t = e^{p_t}$ be the price level. By (3),

$$P_{t+1} = e^{p_t + \mu + \sigma\varepsilon_{t+1}} = P_t\, e^{\mu}\, e^{\sigma\varepsilon_{t+1}}.$$

P_t is known at t and ε_{t+1} is independent of it, so

$$E[P_{t+1}|P_t] = P_t\, e^{\mu}\, E\Big[e^{\sigma\varepsilon}\Big]. \tag{19}$$

The last term can be evaluated by completing the square,

$$E\Big[e^{\sigma\varepsilon}\Big] = \int_{-\infty}^{\infty} e^{\sigma x}\, \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx = e^{\sigma^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x-\sigma)^2/2}\, dx = e^{\sigma^2/2},$$

because the remaining integrand is the density of a normal with mean σ and unit variance, which integrates to one. Substituting into (19),

$$E[P_{t+1}|P_t] = P_t\, e^{\mu+\sigma^2/2} = P_t,$$

because $\mu = -\sigma^2/2$.

By the law of iterated expectations and the one-period result, $E[P_{t+\tau}|P_t] = P_t$ for every $\tau \geq 1$.

Present discounted value. If a block yields expected output q_τ in the τ-th year after drilling starts, the expected present value of that output at the date t when drilling starts is

$$E_t\Big[\sum_{\tau\geq 0} \delta^\tau q_\tau P_{t+\tau}\Big] = P_t \sum_{\tau\geq 0} \delta^\tau q_\tau,$$

which is proportional to the current price.

L.2 Proofs of the Propositions

Proof of Proposition 1. By (6), $V_n(P_t; r, Q_\ell, C) = C\, v_n(S/C)$ with $S = (1-r)P_tQ_\ell$, so a winner with cost C and revenue S drills if and only if a winner with a one-dollar cost drills at revenue S/C. It therefore suffices to show the result for v_n. Let $W_n(S) \equiv \delta\, E\, v_{n-1}(SR)$ be the value of waiting with n months left, so that (6) becomes $v_n = \max\{S-1, W_n\}$.

First, I show that more time is worth more. In the last month $v_1(S) = \max\{S-1, 0\} \geq 0 = v_0(S)$. If $v_{n-1} \geq v_{n-2}$ everywhere, then $W_n \geq W_{n-1}$ and $v_n \geq v_{n-1}$. By induction v_n is nondecreasing in n, and $v_n \geq 0$.

Second, the value of waiting rises more slowly than the value of drilling. I show by induction that v_n is nondecreasing in S and that $v_n'(S) \leq 1$ wherever the derivative exists. This holds for v_1. If it holds for v_{n-1}, then

$$W_n'(S) = \delta\, E\Big[R\, v_{n-1}'(SR)\Big] \leq \delta\, E[R] = \delta \leq 1,$$

using $E[R] = E[e^{\mu+\sigma\varepsilon}] = e^{\mu+\sigma^2/2} = 1$ (Appendix L.1). A maximum of two functions whose

slopes are at most one has slope at most one, so the claim holds for v_n.

Third, there is a threshold. The gain from drilling rather than waiting, $S - 1 - W_n(S)$, is continuous in S, because W_n is an expectation of the continuous function v_{n-1}, and by the second step its slope is at least $1 - \delta > 0$. The gain is therefore strictly increasing and unbounded above. At $S = 0$ it equals $-1 - W_n(0) < 0$, so it crosses zero exactly once, and by continuity the winner drills exactly at the revenues from that crossing on, an interval $[e^{y_n^*}, \infty)$, where

$$y_n^* \equiv \ln \min\{S : S - 1 \geq W_n(S)\}$$

is the log of the lowest revenue at which a winner with a one-dollar cost and n months left drills. Since $W_n \geq 0$, drilling requires $S \geq 1$, so $y_n^* \geq 0$. Since $W_1 = 0$, $y_1^* = 0$. Since $W_n \geq W_{n-1}$, the gain is weakly smaller with more months left, so $y_n^* \geq y_{n-1}^*$. By homogeneity a winner with cost $C_{W\ell}$ drills with n months left if and only if $S \geq C_{W\ell}\, e^{y_n^*}$. Substituting revenue (4) into $S_{\ell t} = (1 - r_{W\ell}) P_t Q_\ell \geq C_{W\ell}\, e^{y^*_{T_\ell - t + 1}}$ gives (7). □

Proof of Proposition 2. The entrant's expected payoff from share r is $\Pi(r) = G_\ell(r)\, V_{T_\ell}(\bar{P}_\ell; r, Q_\ell, C)$, so

$$\Pi'(r) = g_\ell(r)\, V_{T_\ell}(\bar{P}_\ell; r, Q_\ell, C) + G_\ell(r)\, \frac{\partial V_{T_\ell}(\bar{P}_\ell; r, Q_\ell, C)}{\partial r}.$$

At an interior maximum $\Pi'(r) = 0$, which rearranges to (8). □

Proof of Proposition 3. In equilibrium the first-order condition (8) holds at every type with $r = \rho_\ell(x)$. By (6) with $C = Q_\ell / x$, $V_{T_\ell}(\bar{P}_\ell; r, Q_\ell, C) = C\, v_{T_\ell}(a(x))$ and $-\partial V_{T_\ell}/\partial r = C\, \bar{P}_\ell\, x\, v'_{T_\ell}(a(x))$. Substituting and using $\bar{P}_\ell\, x = a(x)/(1 - r)$,

$$\frac{v_{T_\ell}\big(a(x)\big)}{a(x)\, v'_{T_\ell}\big(a(x)\big)} = \frac{G_\ell(r)}{(1 - r)\, g_\ell(r)}.$$

Differentiating with respect to x, the left side changes by its derivative in revenue times $a'(x)$ and the right side by its derivative in r times $\rho'_\ell(x)$. The left side is increasing in revenue (Section 5.2) and $\rho'_\ell > 0$, so $a'(x)$ has the sign of the derivative of the right side. Its log is $\ln G_\ell(r) - \ln g_\ell(r) - \ln(1 - r)$, whose derivative is

$$\frac{1}{1 - r} + \frac{d}{dr} \ln \frac{G_\ell(r)}{g_\ell(r)}.$$

If G_ℓ is log-concave, g_ℓ / G_ℓ is nonincreasing, so G_ℓ / g_ℓ is nondecreasing and both terms are positive. □